

The block below carries no module guard, so docstrip copies it into EVERY generated file. That is what we want for the TeX files and wrong for the latexmkrc, which is Perl and does not understand lines opening with two percent signs – hence the negated guard around it.

The guard lines themselves are hidden behind a false conditional. docstrip reads a guard line either way, but ltxdoc, which typesets this file, would otherwise set the literal text of the guard as a stray paragraph in the body of the manual.

Nothing in this area may name a TeX conditional, not even in prose: outside the meta-comment at the top of the file, every documentation line here IS LaTeX source and is executed, and a conditional named in passing opens or closes a real one. That is not hypothetical – an earlier draft of this very comment swallowed the rest of the manual.

Three conditional primitives are deliberately absent from the lists below: the ones that close or branch a conditional. `\DoNotIndex` scans its argument with TeX’s own conditional machinery live, so a bare one of them closes the wrong conditional. The manual ended its run with “Too many }’s” and exited with status 1, which failed the docs step of the harness even though the PDF had been written. Leaving them out of the lists means they get indexed, which is harmless. Note that they cannot be named here either, not even inside . . . shorthand split across a line.

The coppe document class

Version 4.0

Geraldo Xexéo Vicente H. F. Batista
George O. Ainsworth Jr. Eduardo Mangeli

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a WhatsApp group where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Folha adicional: Coleta CAPES

Since August 2026 every thesis and dissertation must carry a *folha adicional* immediately after the folha de rosto, holding the fields the CAPES/MEC collection requires plus the ficha catalogafica (SiBI manual, 9th rev. ed. 2026, 3.1.2.1.2

and Annex H). The sheet is filled in by the graduate programme, and is neither counted nor numbered.

The macros below feed that sheet. They are institutional data, so the sheet is always typeset in Portuguese regardless of the main language of the work, as the COPPE norm requires of institutional identity.

`\areaconcentracao` The area of concentration, e.g. `\areaconcentracao{Engenharia de Sistemas e Computacao}`. It appears on the folha de rosto, on the folha de aprovacao and on the folha adicional.

`\linhapesquisa` The research line, required by the CAPES model of the folha de rosto.

`\tipoproducao` The kind of intellectual output, one of `bibliografica`, `artistica`, `tecnologica` or `tecnica`. Anything else raises a warning.

`\projetovinculado` Whether the work belongs to a research project (`sim` or `nao`) and, if so, its

`\nomeprojeto` name.

`\agenciafomento` A funding agency, given as full name and acronym; repeat once per agency:

```

\agenciafomento{Conselho Nacional de Desenvolvimento
                  Cientifico e Tecnologico}{CNPq}
\agenciafomento{Fundacao Carlos Chagas Filho de Amparo a
                  Pesquisa do Estado do Rio de Janeiro}{FAPERJ}

```

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one

`foreignabstract (env.)` page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.

`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which

`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is

`\printloabbreviations` provided, it will used as sort key. This was later, in 2024, implemented also for

`\printlosymbols`

`\abbrev`, otherwise `Symbol`, or `Abreviation` will be used as sort key, what can result in an undesirable order if it contains $\LaTeX$ commands, mathematical symbols, or mix of uppercase and lowercase. `MakeIndex` needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain `MakeIndex` input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

`COPPE $\TeX$` produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no `BIB $\TeX$` `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` `BIB $\TeX$` styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib} % in the preamble
...
\printbibliography           % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LaTeX`, `biber`, and twice again `LaTeX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic @book (@livro); @article (@artigo); @artigodejornal (newspaper); @incollection (@partedelivro); @inbook (@emlivro); @inreference (@verbete); @proceedings (@anais); @inproceedings (@trabalhodeevento); @thesis (@tese/@dissertacao, with the presets @dissertacaomestrado, @tesedoutorado, @tesephD, @tcc); @report (@relatorio); @periodical (@periodico); @manual; @misc (@diverso); @online.

Special (NBR 6023 §4.2.5–4.2.14) @patent (@patente); @legislation (@legislacao/@atonormat @jurisdiction (@jurisprudencia, with a relator field); @legal (@documentojuridico); @standard (@norma); @music (@partitura); @map (@mapa); @image (@iconografico); @artwork (@obradearte); @video (@audiovisual); @movie (@filme); @letter (@correspondencia); @dataset (@conjuntodedados).

Games and TV @game (@jogo); @boardgame (@jogodetabuleiro); @videogame (@jogoeletronico); @tvshow (@programadetv).

When an entry has no author or editor, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard L^AT_EX command used before all appendices. COPPET_{EX} introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer's *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian babel), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Computer Modern (padrão L^AT_EX)*. Redefine if you loaded `lmodern`, `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPE_TE_X class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is **brazilian**.

- **brazilian** and **english** are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- **spanish, french, italian** are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.

- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein` / `\brazilianabstract` API, and how to create a new language pack.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the `abstract` environment. Picked by the class option (`brazilian` by default; one of `english`, `spanish`, `french`, `italian` also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the `foreignabstract` environment. `english` by default for every main language; switches automatically to `brazilian` when `main = english`.

third optional. Loaded on demand with `\usecoppelanguage{<lang>}` in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

`\titlein`
`brazilianabstract (env.)`
`\usecoppelanguage`

Two pieces of user-facing API come with the multilingual model:

- `\titlein{<lang>}{<text>}` records the title in any language. `\title{...}` still writes the Brazilian-Portuguese title and `\foreigntitle{...}` still writes the English one (back-compat); for a main language outside the `brazilian/english` pair, also call `\titlein{<lang>}{...}`. The cover and folha de rosto print whichever title matches `\coppe@mainlang`.
- The `brazilianabstract` environment is the optional third abstract used when the main language is neither `brazilian` nor `english` – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in `\begin{otherlanguage}{brazilian}` and uses the Portuguese template strings. A document with `main = brazilian` can simply keep using `abstract`; `brazilianabstract` is redundant then.
- `\usecoppelanguage{<lang>}` loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings `universityname`, `cityname`, `statename`, `countryname`, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to `coppe.cls` (or anywhere on the TEXINPUTS path):

`coppe-lang-<lang>.def` class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for `brazilian/english` in `coppe.cls` (`appendix`, `annex`, `frame`, `listframe`, `source`, `program`, `listprogram`, `references`, `in:`, `art`, `platform`, `directedby`, `producedby`, `listabbreviation`, `listsymbol`, `glossary`, `advisor`, `advisors`, `dept`, `approvedmale`, `approvedfemale`, `universityname`, `cityname`, `statename`, `countryname`, `monthname1...monthname12`, `algorithm2eopt`, `babelname`). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the `biblatex` mapping is registered after the bibliography engine is loaded by `coppe.cls` (which happens *after* the language pack is read).

`<lang>-coppe.lbx` `biblatex` localization strings. `\InheritBibliographyExtras{<lang>}` then `\DeclareBibliographyStrings` for the keys `availablefrom`, `urlseen`, `in`, `editor`, `editors`, `depositor`, `coppescale`, `mscdiss`, `dscthesis`, `phdthesis`, `tcc`. The shipped `spanish-coppe.lbx`, `french-coppe.lbx` and `italian-coppe.lbx` are the reference templates to copy from.

Tip: the `algorithm2eopt` key holds whatever `algorithm2e` calls the language. Mind the spelling: `algorithm2e` ships `portuguese`, `english`, `spanish`, `french` and `italiano` – not `italian`.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T'\i tulo em portuguese} % Brazilian abstract slot
  \foreigntitle{English title} % English abstract slot
  \titlein{spanish}{T'\i tulo en espa~nol} % cover/folha
  \author{Juan}{P\erez}
  \advisor{Prof.}{Mar'\i a}{Garc'\i a}{D.Sc.}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
```

```

\begin{abstract}          Resumen principal en espa\~nol.    \end{abstract}
\begin{foreignabstract} English foreign abstract.           \end{foreignabstract}
\begin{brazilianabstract} Resumo em portugu\~es para a banca. \end{brazilianabstract}
\tableofcontents
\mainmatter
\chapter{Introducci\~on} Cuerpo en espa\~nol\dots
\end{document}

```

6 Files in your project folder

In a normal install where CoppeTeX is part of your `TEXINPUTS` path (e.g. inside the local `texmf` tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```

\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}

```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
<code>brazilian</code> (default)	— (built into <code>coppe.cls</code>)
<code>english</code>	— (built into <code>coppe.cls</code>)
<code>spanish</code>	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
<code>french</code>	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
<code>italian</code>	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Required L^AT_EX packages

The class is built on the standard `book` class (`12pt`, `a4paper`, `oneside`) and brings in the stock CTAN packages listed below with `\RequirePackage`. On a full T_EX Live or MiK_TE_X install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are already in scope. You do *not* need to `\usepackage` any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

`iftex` detects the running engine, so the class can branch its setup between `pdfTEX` and the Unicode engines (LuaT_EX/XeT_EX).

`inputenc (utf8)` UTF-8 input; loaded *only under pdfT_EX* (the Unicode engines read UTF-8 natively).

`fontenc (T1)` T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdfTeX*. Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 one-sided margins (3/3/2/2 cm) and header height.

`fancyhdr` running headers and footers.

`titlesec` chapter and section title formatting.

`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.

`tabularx` tables with automatic-width columns.

`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).

`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.

`float` float placement control (e.g. the `[H]` specifier).

`caption` caption typography (the ABNT caption-on-top layout).

`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.

`fancyvrb` extended verbatim.

`algorithm2e` (`linesnumbered`, `lined`, `algochapter`, `ruled`) typeset algorithms; the `natural-language` option tracks the main language.

Bibliography, quotations and language

`babel` multilingual typesetting; the language options are computed from the main and foreign language slots.

`csquotes` context-sensitive quotation marks (the recommended companion of `biblatex`).

`biblatex` (`backend=biber`) the bibliography engine and the ABNT `coppe` styles (`coppe` or, with the `numbers` option, `coppe-numeric`). The backend is `biber`, not `bibTeX`.

Hyperlinks

`hyperref` clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

`etoolbox` e-TeX programming toolkit.

`ltxcmds` low-level command helpers (`\ltx@ifpackageloaded, \dots`).

`ifthen` the `\ifthenelse` conditional.

`hyphenat` hyphenation control (`\nohyphens`, used on the cover).

`lastpage` `\pageref{LastPage}` for the cataloging page count.

`hologo` TeX-family logos (`\hologo{LaTeX}, \dots`).

`xcolor` colour support (used by `listings` and `hyperref`).

8 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdfL^AT_EX will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (amssymb vs. unicode-math). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under pdfL^AT_EX: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (LuaL^AT_EX, XeL^AT_EX). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math`

nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

longquote Quotation To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

9 A simple example

```
1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating} % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz} % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 \usepackage{unicode-math}
11 \setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe (v4.0) já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib}% base de referências (biblatex/biber)
20
21 \makelosymbols
22 \makeloabbreviations
23 % Algumas coisas opcionais que não cabem no coppetex.sty
24 \renewcommand{\familydefault}{\sfdefault} % para usar fontes SEM serifa no default
25 \renewcommand{\familydefault}{\rmdefault} % para usar fontes COM serifa no default
26
27
28 \begin{document}
```

```

29 \title{Título da Tese}
30 \foreigntitle{Thesis Title}
31 \author{Nome do Autor}{Sobrenome}
32 \advisor[UFRJ]{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
33 \advisor[UFRJ]{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
34 % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
35 % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
36 % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
37 % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
38
39 \examiner[UFRJ]{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
40 \examiner[UFRJ]{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
41 \examiner[UFRJ]{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
42 \examiner[UNICAMP]{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
43 \examiner[USP]{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
44 \department{PESC}
45 \date{06}{2026}
46
47 \keyword{Primeira palavra-chave}
48 \keyword{Segunda palavra-chave}
49 \keyword{Terceira palavra-chave}
50
51 % Dados da folha adicional (Coleta CAPES), obrigatoria desde agosto de 2026.
52 % A folha e preenchida pelo Programa; estes campos apenas a compoem.
53 \areaconcentracao{Engenharia de Sistemas e Computa\c c\~ao}
54 \linhapesquisa{Engenharia de Dados e Conhecimento}
55 \tipoproducao{bibliografica} % bibliografica | artistica | tecnologica | tecnica
56 \projetovinculado{sim} % sim | nao
57 \nomeprojeto{Nome do Projeto de Pesquisa ao qual o trabalho se vincula}
58 \agenciafomento{Conselho Nacional de Desenvolvimento Cientifico e
59 Tecnologico}{CNPq}
60 \agenciafomento{Fundacao Carlos Chagas Filho de Amparo a Pesquisa do
61 Estado do Rio de Janeiro}{FAPERJ}
62 % A ficha oficial vem de fichacatalografica.sibi.ufrj.br; tendo o arquivo,
63 % descomente a linha abaixo e a moldura vazia da folha adicional some.
64 % \fichacatalografica{ficha.pdf}
65
66 \maketitle
67
68 \frontmatter
69 \dedication{A alguém cujo valor é digno desta dedicatória.}
70
71 \chapter*{Agradecimentos}
72
73 Gostaria de agradecer a todos os que contribu\iram para este trabalho:
74 orientadores, colegas, familiares e \as ag\encias de fomento. Lorem ipsum
75 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
76 ut labore et dolore magna aliqua.
77
78 \begin{abstract}
79
80 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
81 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
82 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation

```



```

83 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
84 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
85 pariat. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
86 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
87 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
88 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
89 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
90 aspernatur aut odit aut fugit.
91
92 \end{abstract}
93
94 \begin{foreignabstract}
95
96 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
97 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
98 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
99 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
100 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
101 pariat. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
102 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
103 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
104 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
105 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
106 aspernatur aut odit aut fugit.
107
108 \end{foreignabstract}
109
110 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
111 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
112 \listoffigures
113 \listoftables
114 \listofquadros
115 \listofprogramas
116 \listofalgorithms
117 \printloabbreviations
118 \printlosymbols
119 \tableofcontents
120
121 \mainmatter
122 \chapter{Introdução}
123
124 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
125
126 A classe \verb|coppe| foi criada originalmente por Vicente Helano e George Ainsworth, poré
127 Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M
128
129 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação
130
131 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.com/CO
132
133 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
134
135 Há um grupo de WhatsApp para perguntas e discussões do estilo: \href{https://github.com/CO
136

```

137 Bugs devem ser reportados na \href{https://github.com/COPPE-UFRJ/CoppeTeX/issues}{página d
138
139 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
140
141 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \ para a A
142
143 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma
144
145 Este documento não substitui, mas complementa, o documento que descreve a classe.
146
147
148
149 \chapter{Configurações Iniciais}
150
151 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
152
153 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
154
155 Recomendo ler o documento “‘The \verb*|coppe| document class” para entender melhor todas
156
157 \section{Linguagem principal do texto}
158
159 Essa classe considera que o texto principal está em português e algumas partes específicas
160 \begin{itemize}
161 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.
162 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
163 \end{itemize}
164
165 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe \CoppeT
166
167
168 \section{Por que usar o \LaTeX}
169
170 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess
171
172 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p
173
174 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
175
176 \section{Como e onde usar o \LaTeX}
177
178 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambiente
179 \begin{enumerate}
180 \item Na sua máquina, instalando uma versão completa como o \hologo{MiKTeX}, típico do Win
181 \item Usar um ambiente na rede, como o Overleaf.
182 \end{enumerate}
183
184 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b
185
186
187 \chapter{Algumas Regras da COPPE}
188
189 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei
190

191 É imprescindível definir os símbolos, tal como o
192 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
193 $\text{\syml}{\mathbb{R}}$ {Conjunto dos números reais}
194 $\text{\syml}{\emptyset}$ {Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
195
196 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \ve
197
198 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice
199
200 \section{Citações}
201 \label{sec:citacoes}
202
203 Citações curtas podem ser feitas \enquote{o comando quote} ou direto com “duas crases e do
204
205 \begin{longquote}
206 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
207 feita com o ambiente \verb*|longquote| The primary objective of this
208 investigation was to determine the feasibility of detecting corrosion in
209 aluminum Naval aircraft components with neutron radiographic interrogation
210 and the use of standard corrosion penetrameters. Secondary objectives
211 included the determination of the effect of object thickness on image quality,
212 the defining of minimum levels of detectability and a preliminary investigation
213 of a means whereby the degree of corrosion could be quantified with neutron
214 radiographic data. \citep{article-example}
215 \end{longquote}
216
217 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \verb*|na
218
219 Em todo caso, \textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em
220
221 Quando se cita uma obra por meio de outra (\textit{apud}), pela ABNT entra na lista de ref
222
223 \chapter{Floats}
224
225 Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \texti
226
227 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm
228
229 \textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração
230
231 Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao
232
233
234 \section{Tabelas e Figuras Padrão}
235
236 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.
237
238 \begin{table}[ht]
239 \centering % Centraliza a tabela
240 \caption{Exemplo de Tabela de Números}
241 \label{tab:exemplo_numeros}
242 \begin{tabular}{ccc} % Define a quantidade de colunas
243 \hline % Linha superior
244 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \textbf{\% Cabeçalhos}

```

245 \hline % Linha média
246 1 & 2 & 3 \\ % Primeira linha de dados
247 \hline
248 4 & 5 & 6 \\ % Segunda linha de dados
249 \hline
250 7 & 8 & 9 \\ % Terceira linha de dados
251 \hline
252 10 & 11 & 12 \\ % Quarta linha de dados
253 \hline % Linha inferior
254 \end{tabular}
255 \fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatoria, abaixo do float
256 \end{table}
257
258
259
260 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
261
262 \begin{figure}[ht]
263 \centering % Centraliza a figura
264 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
265 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
266 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
267 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
268 \end{figure}
269
270 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
271
272 \begin{quadro}[ht]
273 \centering
274 \caption{Exemplo de Quadro}
275 \label{qua:exemplo}
276 \begin{tabular}{|l|l|}
277 \hline
278 \textbf{Ilustração} & \textbf{Delimitação} \\
279 \hline
280 Tabela & Linhas apenas no topo e na base. \\
281 \hline
282 Quadro & Linhas em todas as bordas. \\
283 \hline
284 \end{tabular}
285 \fonte{Elabora\c{c}\~ao pr\'opria.}
286 \end{quadro}
287
288
289
290
291
292 \section{Tabelas mais elegantes}
293
294 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
295
296 \begin{table}[ht]
297 \centering % Centraliza a tabela
298 \caption{Exemplo de Tabela de Números mais elegantes}

```

```

299 \label{tab:exemplo_numerosbom}
300 \begin{tabular}{ccc} % Define a quantidade de colunas
301 \toprule % Linha superior
302 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
303 \midrule % Linha média
304 1 & 2 & 3 \\ % Primeira linha de dados
305 4 & 5 & 6 \\ % Segunda linha de dados
306 7 & 8 & 9 \\ % Terceira linha de dados
307 10 & 11 & 12 \\ % Quarta linha de dados
308 \bottomrule % Linha inferior
309 \end{tabular}
310 \fonte{Elabora\c{c}\~ao pr\'opria.}
311 \end{table}
312
313 \section{Tabelas Longas ou Largas}
314
315 Se sua tabela é muito longa ou larga, existem várias opções.
316 \begin{itemize}
317     \item alterar o tamanho da letra
318     \item Usar o longtable
319     \item rodar a tabela, fazendo ela em \textit{landscape}
320     \item fazer a tabela dentro de um minibox
321 \end{itemize}
322
323
324 \subsection{Tabelas largas demais}
325
326 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
327
328 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
329
330 \begin{table}[ht]
331 \centering % Centraliza a tabela
332 \caption{Exemplo de Tabela Larga com Fonte Menor}
333 \label{tab:tabela_largafns}
334 \footnotesize % Aplica uma fonte menor para a tabela
335 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
336 \toprule
337 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} & \textbf{Col. 4} & \textbf{Col. 5} & \textbf{Col. 6} & \textbf{Col. 7} \\
338 \midrule
339 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
340 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
341 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
342 \bottomrule
343 \end{tabular}
344 \fonte{Elabora\c{c}\~ao pr\'opria.}
345 \end{table}
346
347 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
348
349 \begin{table}[ht]
350 \centering
351 \caption{Exemplo de Tabela Redimensionada}
352 \label{tab:examplerb}

```

```

353 \resizebox{\textwidth}{!}{%
354 \begin{tabular}{llll}
355 \toprule
356 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
357 \midrule
358 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
359 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
360 \bottomrule
361 \end{tabular}%
362 }
363 \fonte{Elabora\c{c}\~ao pr\'opria.}
364 \end{table}
365
366
367 Para rodar uma tabela muito larga em 90 graus no LaTeX, voc\ea pode usar o pacote \verb*|rotat
368
369 Aqui est\aa um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
370
371 \begin{sidewaystable}
372 \centering
373 \caption{Sua Legenda Aqui}
374 \label{tab:sua_tabela}
375 \begin{tabular}{lll}
376 \toprule
377 Coluna 1 & Coluna 2 & Coluna 3 \\
378 \midrule
379 Item 1 & Item 2 & Item 3 \\
380 Item 4 & Item 5 & Item 6 \\
381 \bottomrule
382 \end{tabular}
383 \fonte{Elabora\c{c}\~ao pr\'opria.}
384 \end{sidewaystable}
385
386 Se a tabela for muito longa, o ambiente \verb|longtable| \e o ideal. Ele fornece comandos par
387
388 % Exemplo de tabela longa que se estende por v\arias p\aginas
389 \begin{longtable}{|c|c|c|}
390 % primeiro cabe\c alho (\e o caption)
391 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
392 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
393 \endfirsthead
394 % cabe\c alho normal
395 \multicolumn{3}{c}%
396 {\tablelongname\ \thetable} -- continua\c ao da p\agina anterior} \\
397 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
398 \endhead
399 % p\e normal
400 \hline \multicolumn{3}{|r|}{Continua na pr\oxima p\agina} \\ \hline
401 \endfoot
402 \hline
403 % \u ltimo p\e
404 \multicolumn{3}{|r|}{Continua na pr\oxima p\agina} \\
405 \hline \hline
406 \endlastfoot

```

```

407
408 % Conteúdo da tabela
409 1 & 2 & 3 \\
410 4 & 5 & 6 \\
411 1 & 2 & 3 \\
412 4 & 5 & 6 \\
413 1 & 2 & 3 \\
414 4 & 5 & 6 \\
415 1 & 2 & 3 \\
416 4 & 5 & 6 \\
417 1 & 2 & 3 \\
418 4 & 5 & 6 \\
419 1 & 2 & 3 \\
420 4 & 5 & 6 \\
421 1 & 2 & 3 \\
422 4 & 5 & 6 \\
423 1 & 2 & 3 \\
424 4 & 5 & 6 \\
425 1 & 2 & 3 \\
426 1 & 2 & 3 \\
427 4 & 5 & 6 \\
428 1 & 2 & 3 \\
429 4 & 5 & 6 \\
430 1 & 2 & 3 \\
431 4 & 5 & 6 \\
432 1 & 2 & 3 \\
433 4 & 5 & 6 \\
434 1 & 2 & 3 \\
435 4 & 5 & 6 \\
436 1 & 2 & 3 \\
437 4 & 5 & 6 \\
438 1 & 2 & 3 \\
439 4 & 5 & 6 \\
440 1 & 2 & 3 \\
441 4 & 5 & 6 \\
442 1 & 2 & 3 \\
443 4 & 5 & 6 \\
444 1 & 2 & 3 \\
445 4 & 5 & 6 \\
446 1 & 2 & 3 \\
447 4 & 5 & 6 \\
448 1 & 2 & 3 \\
449 4 & 5 & 6 \\1 & 2 & 3 \\
450 4 & 5 & 6 \\
451 1 & 2 & 3 \\
452 4 & 5 & 6 \\
453 1 & 2 & 3 \\
454 1 & 2 & 3 \\
455 4 & 5 & 6 \\
456 1 & 2 & 3 \\
457 4 & 5 & 6 \\
458 1 & 2 & 3 \\
459 4 & 5 & 6 \\
460 1 & 2 & 3 \\

```

461 4 & 5 & 6 \\
 462 1 & 2 & 3 \\
 463 4 & 5 & 6 \\
 464 1 & 2 & 3 \\
 465 4 & 5 & 6 \\
 466 1 & 2 & 3 \\
 467 4 & 5 & 6 \\
 468 1 & 2 & 3 \\
 469 4 & 5 & 6 \\
 470 1 & 2 & 3 \\
 471 4 & 5 & 6 \\
 472 1 & 2 & 3 \\
 473 4 & 5 & 6 \\
 474 1 & 2 & 3 \\
 475 4 & 5 & 6 \\
 476 1 & 2 & 3 \\
 477 4 & 5 & 6 \\
 478 1 & 2 & 3 \\
 479 4 & 5 & 6 \\
 480 1 & 2 & 3 \\
 481 4 & 5 & 6 \\
 482 1 & 2 & 3 \\
 483 4 & 5 & 6 \\
 484 1 & 2 & 3 \\
 485 4 & 5 & 6 \\
 486 1 & 2 & 3 \\
 487 4 & 5 & 6 \\
 488 1 & 2 & 3 \\
 489 4 & 5 & 6 \\
 490 1 & 2 & 3 \\
 491 4 & 5 & 6 \\
 492 1 & 2 & 3 \\
 493 4 & 5 & 6 \\
 494 1 & 2 & 3 \\
 495 4 & 5 & 6 \\
 496 1 & 2 & 3 \\
 497 4 & 5 & 6 \\
 498 1 & 2 & 3 \\
 499 4 & 5 & 6 \\
 500 1 & 2 & 3 \\
 501 4 & 5 & 6 \\
 502 1 & 2 & 3 \\
 503 4 & 5 & 6 \\
 504 1 & 2 & 3 \\
 505 4 & 5 & 6 \\
 506 1 & 2 & 3 \\
 507 4 & 5 & 6 \\
 508 1 & 2 & 3 \\
 509 4 & 5 & 6 \\
 510 1 & 2 & 3 \\
 511 4 & 5 & 6 \\
 512 1 & 2 & 3 \\
 513 4 & 5 & 6 \\
 514 1 & 2 & 3 \\


```

515 4 & 5 & 6 \\
516
517 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
518 \end{longtable}
519 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
520
521 \section{Subfiguras (subcaption)}
522
523 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
524
525 \begin{figure}[ht]
526   \centering
527   \begin{subfigure}[b]{0.42\textwidth}
528     \centering\includegraphics[width=\linewidth]{example-image-a}
529     \caption{Primeira vista}\label{fig:sub-a}
530   \end{subfigure}
531   \hfill
532   \begin{subfigure}[b]{0.42\textwidth}
533     \centering\includegraphics[width=\linewidth]{example-image-b}
534     \caption{Segunda vista}\label{fig:sub-b}
535   \end{subfigure}
536   \caption{Figura com duas subfiguras}
537   \label{fig:subs}
538   \fonte{Elabora\c{c}\~ao pr\'opria.}
539 \end{figure}
540
541 Referencie a figura inteira com \verb|\autoref{fig:subs}|. Cada parte tem
542 rótulo próprio: \verb|\autoref{fig:sub-a}| leva à \autoref{fig:sub-a}.
543
544 \section{Imagens (graphicx)}
545
546 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
547 \begin{itemize}
548   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
549   \item \verb|[angle=90]| --- rotação;
550   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, direi}
551 \end{itemize}
552 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt{
553
554 \section{Introdução ao tikz}
555
556 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
557
558 \begin{figure}[ht]
559   \centering
560   \begin{tikzpicture}
561     \node[draw, rounded corners] (a) at (0,0) {Início};
562     \node[draw, rounded corners] (b) at (4,0) {Fim};
563     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
564   \end{tikzpicture}
565   \caption{Diagrama simples em tikz}
566   \label{fig:tikz}
567   \fonte{Elabora\c{c}\~ao pr\'opria.}
568 \end{figure}

```

```

569
570 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|ti
571
572 \section{Float ou direto no texto?}
573
574 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \L
575
576 \chapter{Using BibLaTeX}
577 \label{cap:biblatex}
578
579 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
580
581 \textbf{É} obrigatório compilar com \texttt{biber} (e não com \hologo{BibTeX}). A sequênci
582
583 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
584
585 \begin{table}[h]
586 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr~ao
587 \texttt{\textbackslash cite} do \LaTeX\ e
588 o comando \texttt{\textbackslash citep},
589 disponível pela opção \texttt{natbib} do BibLaTeX.}
590 \label{tab:citation}
591 \centering
592 {\footnotesize
593 \begin{tabular}{|c|c|c|}
594 \hline
595 Tipo da Publicação & \verb|\cite| & \verb|\citep|\\
596 \hline
597 Livro & \cite{book-example} & \citep{book-example}\\
598 Artigo & \cite{article-example} & \citep{article-example}\\
599 Relatório & \cite{techreport-example} & \citep{techreport-example}\\
600 Relatório & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\
601 Anais de Congresso & \cite{inproceedings-example} &
602 \citep{inproceedings-example}\\
603 Séries & \cite{incollection-example} & \citep{incollection-example}\\
604 Em Livro & \cite{inbook-example} & \citep{inbook-example}\\
605 Dissertação de mestrado & \cite{mastersthesis-example} &
606 \citep{mastersthesis-example}\\
607 Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\
608 \hline
609 \end{tabular}}
610 \fonte{Elabora{\c c}\~ao pr'opria.}
611 \end{table}
612
613 \begin{table}[h]
614 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr~ao
615 \texttt{\textbackslash cite} do \LaTeX\ e
616 o comando \texttt{\textbackslash citet},
617 disponível pela opção \texttt{natbib} do BibLaTeX. Além disso, usando o booktabs.}
618 \label{tab:citation1}
619 \centering
620 {\footnotesize
621 \begin{tabular}{ccc}
622 \toprule

```

```

623 Tipo da Publicação & \verb|\cite| & \verb|\citet|\
624 \midrule
625 Livro & \cite{book-example} & \citet{book-example}\
626 Artigo & \cite{article-example} & \citet{article-example}\
627 Relatório & \cite{techreport-example} & \citet{techreport-example}\
628 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\
629 Anais de Congresso & \cite{inproceedings-example} &
630 \citet{inproceedings-example}\
631 Séries & \cite{incollection-example} & \citet{incollection-example}\
632 Em Livro & \cite{inbook-example} & \citet{inbook-example}\
633 Dissertação de mestrado & \cite{mastersthesis-example} &
634 \citet{mastersthesis-example}\
635 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\
636 \bottomrule
637 \end{tabular}}
638 \fonte{Elabora\c{c}\~ao pr\`opria.}
639 \end{table}
640
641 \section{Comandos de citação}
642
643 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
644
645 \subsection{Parâmetro opcional: página e “tradução nossa”}
646
647 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser
648
649 \subsection{Citação de citação (\emph{apud})}
650
651 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citeapud| (demonstra
652
653 \subsection{Múltiplas bibliografias}
654
655 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|
656
657 \section{Tipos de entrada bibliográfica}
658
659 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinôni
660
661 \subsection{Tipos acadêmicos principais}
662 \begin{itemize}
663 \item Livro / monografia --- \verb|@book| (\verb|@livro|)
664 \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \v
665 \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
666 \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
667 \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
668 \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periód
669 \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sit
670 \end{itemize}
671
672 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
673 \begin{itemize}
674 \item Patente --- \verb|@patent| (\verb|@patente|)
675 \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato
676 \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\v

```

```

677 \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
678 \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb|
679 \end{itemize}
680
681 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
682 \begin{itemize}
683 \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@j
684 \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadettv|, \verb|@serietv|)
685 \end{itemize}
686
687 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
688
689 \chapter{Alguns outros exemplo úteis}
690
691 \begin{tcolorbox}[title=Meu Textbox]
692 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
693 \end{tcolorbox}
694
695 \begin{tcolorbox}
696 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
697 \end{tcolorbox}
698
699 \begin{figure}[ht]
700 \centering
701 \begin{tikzpicture}
702 \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{
703 \begin{scope}[x={(image.south east)},y={(image.north west)}]
704 % Definindo o textbox dentro da figura
705 \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=
706 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
707 Este textbox fala sobre como inserir um textbox dentro de uma figura usa
708 \end{tcolorbox}
709 };
710 \end{scope}
711 \end{tikzpicture}
712 \caption{Figura com Textbox}
713 \label{fig:figura_com_textbox1}
714 \fonte{Elabora\c{c}\~ao pr\'opria.}
715 \end{figure}
716
717
718 \begin{figure}[ht]
719 \centering
720 \begin{tcolorbox}
721 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
722 \end{tcolorbox}
723 \caption{Figura com Textbox simples}
724 \label{fig:figura_com_textbox}
725 \fonte{Elabora\c{c}\~ao pr\'opria.}
726 \end{figure}
727
728 \section{Listagens de código e algoritmos}
729
730 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg

```

```

731
732 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
733 def saudacao(nome):
734     # imprime uma saudação acentuada
735     print(f"Olá, {nome}!")
736
737 saudacao("COPPE")
738 \end{python}
739 \source{Elaboração própria.}
740
741 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
742
743 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
744 Olá, COPPE!
745 \end{Verbatim}
746
747 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|.
748 Há também \verb|\pythoninline| e \verb|\javainline| para código no meio do
749 texto, e o ambiente \verb|Verbatim| (com as opções acima) para a saída de programas. Para al
750
751 \begin{algorithm}[H]
752 \caption{Soma dos elementos de um vetor}
753 \label{alg:soma}
754 \Dados{vetor  $v$  com  $n$  elementos}
755 \Resultado{soma  $s$  dos elementos de  $v$ }
756  $s \leftarrow 0$ ;
757 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
758      $s \leftarrow s + v[i]$ ;
759 }
760 \Retorna  $s$ ;
761 \end{algorithm}
762
763 \chapter{Escrevendo matemática}
764
765 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carr
766
767 \section{Alinhamento (align)}
768 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
769 \begin{align}
770     f(x) &= (x+1)^2 \\
771         &= x^2 + 2x + 1 .
772 \end{align}
773
774 \section{Matrizes (matrix)}
775 O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
776 \begin{equation}
777     A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
778     \quad
779     \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
780 \end{equation}
781
782 \section{Subequações (subequations)}
783 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
784 \begin{subequations}

```

```

785 \begin{align}
786   a &= b + c , \\
787   d &= e - f .
788 \end{align}
789 \end{subequations}
790
791 \section{Estruturas alinhadas (array)}
792 0 \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
793 \begin{equation}
794   |x| = \left\{ \begin{array}{ll}
795     x, & \text{se } x \geq 0 , \\
796     -x, & \text{caso contrário.}
797   \end{array} \right.
798 \end{equation}
799
800 \section{Operadores (\textbackslash DeclareMathOperator)}
801 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
802 \begin{verbatim}
803 \usepackage{amsmath}
804 \DeclareMathOperator{\RMSE}{RMSE}
805 \DeclareMathOperator{\diag}{diag}
806 \end{verbatim}
807 Assim:
808 \begin{itemize}
809   \item \verb|\RMSE(y,\hat{y})$| produz  $\text{RMSE}(y,\hat{y})$ ;
810   \item \verb|\diag(\lambda_1,\lambda_2)$| produz
811      $\text{diag}(\lambda_1,\lambda_2)$ .
812 \end{itemize}
813
814 \section{Unicode e LuaLaTeX (unicode-math)}
815 \label{sec:unicode-math}
816 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss
817 \begin{verbatim}
818 \usepackage{amsmath} % a classe já carrega
819 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
820 \setmathfont{Latin Modern Math} % ou STIX Two Math
821 \end{verbatim}
822 0 código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
823
824 \chapter{Referências cruzadas, notas e índices}
825 \label{cap:refs}
826
827 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
828
829 \section{Rótulos e referências cruzadas}
830
831 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
832
833 \section{Notas de rodapé}
834
835 Use \verb|\footnote{texto}| \footnote{Como esta nota, que a classe imprime em espaçamento s
836
837 \section{hyperref e marcadores}
838

```

```

839 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|ref|/\verb|autoref
840
841 \section{URLs}
842
843 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
844 use \verb|\href{https://exemplo.org}{texto}|. Evite digitar URLs como texto comum --- elas n
845
846 \section{Índice remissivo}
847
848 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
849
850 \section{Traduções (NBR 10520)}
851
852 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
853
854 \chapter{Truques de \LaTeX}
855
856 \section{Comentários mágicos (\texttt{\% !TeX})}
857 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
858 \begin{verbatim}
859 % !TeX program = lualatex
860 % !TeX encoding = UTF-8
861 % !TeX spellcheck = pt_BR
862 % !BIB program = biber
863 \end{verbatim}
864 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
865
866 \section{Espaço engolido}
867 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é
868
869 \section{Escopo (grupos)}
870 Chaves \verb|{...}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças
871
872 \section{Descarregar figuras: \texttt{\textbackslash clearpage} × \texttt{\textbackslash n
873 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
874
875 \section{\texttt{latexmkrc}, perl e Overleaf}
876 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
877
878 \chapter{Método Proposto}
879 \chapter{Resultados e Discussões}
880
881 \section{Algumas Demonstrações}
882
883 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha$
884 \syml[beta]{Beta}{A palavra Beta more e corrigida}
885 \syml[zzbeta]{\beta}{A letra $\beta$ corrigida}
886 \syml[beta]{A palavra beta}
887 \syml[alpha]{A palavra alpha}
888 \syml[alpha]{Alpha}{A palavra Alpha}
889 \syml[zzalpha]{\alpha}{A letra $\alpha$ corrigida}
890 \syml[marco]{Marco}{A palavra Marco corrigida}
891
892 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exempl

```

```

893 \abbrev{GoT}{Game of Thrones}
894 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
895 \abbrev[iot]{IoT}{IoT ordenado como iot}
896 \abbrev[IOT]{IoT}{IoT ordenado como IoT}
897 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
898 \abbrev{IoT}{IoT com ordenação default}
899 \abbrev[ITU]{ITU}{ITU mesmo}
900
901 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
902 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
903
904 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas auto
905
906
907
908
909 \chapter{Conclusões}
910
911 \backmatter
912
913 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
914 % partitura --- declarados via sinônimos em português em example.bib.
915 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
916 \nocite{jurisprudencia-example}
917 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
918 \nocite{jornalcaderno-example,jornal-example}
919 \printbibliography
920
921
922
923 \appendix
924
925 \chapter{Um apêndice}
926
927 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
928
929 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
930
931
932 \chapter{Outro apêndice}
933
934 \annex
935
936
937 \chapter{Um Anexo}
938 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
939
940
941
942 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
943
944 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comand
945
946

```



```

947 \chapter{Intro Anexo}
948
949
950 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
951
952 \end{document}
953
954 \example

```

10 Implementation

10.1 The ‘coppe.cls’ file

```

955 \*class
956 \def\filename{coppe.dtx}
957 \def\fileversion{v4.0}
958 \def\filedate{2026/05/28}
959 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
960 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
961 % Save the class version and date into private macros immediately, before any
962 % required package can redefine the very common \fileversion / \filedate
963 % global names (LaTeX has no convention reserving them for the host class).
964 % \coppe@version / \coppe@date are what user-facing macros (e.g.
965 % \coppetexfinalpage) should reference.
966 \let\coppe@version\fileversion
967 \let\coppe@date\filedate
968 % Document date stored in private slots so \date can set them without ever
969 % touching TeX's \month/\year primitives (which would break \today). They
970 % default to the current system month/year, so a document without \date
971 % still renders sensibly on the cover and folha de rosto.
972 \edef\coppe@docmonth{\the\month}
973 \edef\coppe@docyear{\the\year}
974 \LoadClass[12pt,a4paper,oneside]{book}
975 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
976 % the END of the class (after every class hook is defined), in that order:
977 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
978 % it patches (TOC, refs, lists) are already defined, and biblatex must come
979 % after hyperref so its citation links and back-references work.
980 \RequirePackage{hyphenat}
981 \RequirePackage{lastpage}
982 \RequirePackage{ifthen}
983 \RequirePackage{graphicx}
984 \RequirePackage{tabularx}
985 \RequirePackage{booktabs} % regras de tabela profissionais (toprule/midrule/bottomrule)
986 \RequirePackage{longtable} % tabelas que quebram entre páginas
987 \RequirePackage{etoolbox}
988 \RequirePackage{ltxcmds}
989 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
990 % Render \TeX, \LaTeX and friends via hologo (\CoppTeX is a custom logo, kept).
991 \def\TeX{\hologo{TeX}}
992 \def\LaTeX{\hologo{LaTeX}}
993 \def\LaTeXe{\hologo{LaTeXe}}
994 \def\BibTeX{\hologo{BibTeX}}
995 \def\pdfLaTeX{\hologo{pdfLaTeX}}

```

```

996 \def\pdfTeX{\hologo{pdfTeX}}
997 \def\LuaLaTeX{\hologo{LuaLaTeX}}
998 \def\XeLaTeX{\hologo{XeLaTeX}}
999 \def\MiKTeX{\hologo{MiKTeX}}
1000 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
1001 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
1002 % files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
1003 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
1004 \RequirePackage{iftex}
1005 \ifPDFTeX
1006   \RequirePackage[utf8]{inputenc}
1007   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
1008     Although slower, you should try compiling with LuaLaTeX}}
1009 \fi
1010 % lmodern MUST come with the T1 encoding. Without it, T1 falls back to the EC
1011 % fonts, which exist only as METAFONT bitmaps in most installations, and the
1012 % whole document is typeset in Type3 bitmap fonts. Measured before this change:
1013 % example.pdf carried 87 Type3 fonts against 6 Type1, and coppe.pdf 133 against
1014 % 2. The consequences are not cosmetic:
1015 %
1016 %   * the text renders blurred at any zoom and prints poorly;
1017 %   * copying and searching text in the PDF is unreliable;
1018 %   * PDF/A -- required for deposit by 2.2(d) of the 2026 manual -- is not
1019 %     attainable with bitmap text.
1020 %
1021 % Latin Modern is the vector successor of Computer Modern, metrically and
1022 % visually near-identical, and ships with every current distribution, so the
1023 % page layout does not move.
1024 \RequirePackage{lmodern}
1025 \RequirePackage[T1]{fontenc}
1026 % Math support. amsmath is universal and works on every engine, so the class
1027 % brings it in unconditionally (it must precede hyperref, which is loaded at
1028 % the end of the class). amssymb is the legacy AMS symbol package, built
1029 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
1030 % XeLaTeX) it clashes with unicode-math, which already provides all of its
1031 % symbols through Unicode math codes. So amssymb is loaded ONLY under
1032 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
1033 % unicode-math (the class does not auto-load unicode-math because choosing
1034 % an OpenType math font is a typographic decision that belongs to the author).
1035 \RequirePackage{amsmath}
1036 \ifPDFTeX
1037   \RequirePackage{amssymb}
1038 \else
1039   % Unicode engine: defer the check to end of document. Only warn if the
1040   % user did NOT add either unicode-math (the modern OpenType-math route)
1041   % or amssymb (the legacy route, which they would have loaded manually).
1042   % This avoids nagging authors who already followed the documented
1043   % \usepackage{unicode-math} workflow.
1044   \AtEndDocument{%
1045     \@ifpackageloaded{unicode-math}{\}%
1046     \@ifpackageloaded{amssymb}{\}%
1047     \ClassWarningNoLine{coppe}{%
1048       You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1049       neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak

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```

1050      (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\Message
1051      Add to your preamble:\MessageBreak
1052      \space\space\string\usepackage{unicode-math}\MessageBreak
1053      \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
1054      Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1055      with the OpenType math font setup}}}}
1056 \fi
1057 % Margins per the SiBI manual, 9th rev. ed. (2026), 2.3: left 3cm, top 3cm,
1058 % right 2cm, bottom 2cm -- ONE-SIDED. The 2025 edition also specified verso
1059 % margins (right 3cm / left 2cm) and recommended printing the body on both
1060 % sides; the 2026 revision DELETED the verso margins entirely, following CEPG
1061 % Resolution 246/2023, which ended the printed deposit. A digital-only document
1062 % has no gutter, so the mirrored layout (previously inner=outer=2cm plus a 1cm
1063 % bindingoffset) no longer has any normative basis and is now opt-in via the
1064 % \texttt{twoside} class option, handled after \ProcessOptions below.
1065 %
1066 % headheight=15pt is given here (not patched later) so geometry's own page
1067 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
1068 % height to host the running head, and the standard 12pt default triggers
1069 % its "headheight is too small" warning. Letting geometry know up front
1070 % keeps \topmargin a derived value -- we never touch it by hand.
1071 % (geometry has no \texttt{oneside} key -- only \texttt{twoside}; with the key
1072 % absent it follows the class, which \LoadClass set to one-sided above.)
1073 % headsep is given explicitly to place the folio. 2.7 measures it the way it
1074 % measures the right margin -- to the digits themselves: "a 2cm da borda
1075 % superior, ficando o ultimo algarismo a 2cm da borda direita". The right edge
1076 % already landed at 2.02cm; the top of the digits sat at 1.87cm, too high.
1077 % geometry derives \topmargin from top - 1in - headheight - headsep, so
1078 % shortening headsep lowers the header and leaves the body top untouched:
1079 % topmargin + headheight + headsep is 13.088pt either way, and 3cm is where the
1080 % body still starts. The value is CALIBRATED, not computed -- the folio sits a
1081 % little lower than the arithmetic alone predicts, because fancyhdr's box does
1082 % not put the digits flush with the top of \headheight. Two measured points,
1083 % taken from the rendered page rather than from the PDF's font metrics:
1084 %   headsep 18.07pt -> digits at 1.8669cm   headsep 14.29pt -> 2.0616cm
1085 % which put 2.0000cm at 15.49pt. Measured back on the rendered page, the top of
1086 % the digits and their right edge now come out at the same 2.019cm -- and the
1087 % right margin is 2cm by construction, so that shared 0.019cm is the ruler's
1088 % own bias, not a residual error. Change the body font size or headheight and
1089 % this number has to be measured again, not scaled.
1090 \RequirePackage[a4paper,headheight=15pt,headsep=15.49pt,%
1091 top=3cm,bottom=2cm,left=3cm,right=2cm]{geometry}
1092 % The first paragraph of every section must be indented like any other. LaTeX's
1093 % default is the Anglo-Saxon convention -- no indent after a heading -- which is
1094 % not what Brazilian typography (nor the SiBI manual's own examples) uses.
1095 % indentfirst was mentioned only as a commented suggestion in the example
1096 % preamble, so every document built with this class came out with the first
1097 % paragraph of each section flush left. Loading it in the class fixes it for
1098 % everybody instead of asking each author to remember.
1099 %
1100 % Loaded BEFORE titlesec so titlesec's \@afterindent handling sees the patched
1101 % \@afterindentfalse.
1102 % Long unbreakable tokens -- \verb|...| in running prose, command names, URLs --
1103 % cannot be hyphenated, so TeX overfills the line rather than stretching the

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1104 % spaces. example.tex alone produced 17 overfull boxes this way, the worst 86pt,
1105 % which is the "text does not always respect the margins" a recent review
1106 % reported. \emergencystretch gives TeX a final pass in which it may stretch
1107 % interword spaces instead of running into the margin. It cannot save a single
1108 % token longer than the measure, but it clears the ordinary cases.
1109 \setlength\emergencystretch{3em}
1110 \RequirePackage[indentfirst]
1111 \RequirePackage{titlesec}
1112 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1113 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitolo N").
1114 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1115 % keep "N Title". \appendix and \annex raise the flag (see below).
1116 \newif\if@coppeappendix \@coppeappendixfalse
1117 \titleformat{\chapter}[hang]
1118   {\normalfont\Large\bfseries}
1119   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1120   {1ex}\MakeUppercase}
1121 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1122 \let\coppe@orig@appendix\appendix
1123 \renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
1124 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1125 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1126 % which macro is written (so it is correct at .toc-render time even though the
1127 % appendixname has changed back); the chapter letter is baked in at write time.
1128 \newif\if@coppeannex \@coppeannexfalse
1129 \newcommand\coppe@tocapp[1]{%
1130   \MakeUppercase{\coppestring{appendix}}%
1131   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1132 \newcommand\coppe@tocanx[1]{%
1133   \MakeUppercase{\coppestring{annex}}%
1134   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1135 \def\@chapter[#1]#2{%
1136   \ifnum \c@secnumdepth >\m@ne
1137     \if@mainmatter
1138       \refstepcounter{chapter}%
1139       \typeout{\@chapapp\space\thechapter.}%
1140       \if@coppeappendix
1141         \if@coppeannex
1142           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1143         \else
1144           \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1145         \fi
1146       \else
1147         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1148       \fi
1149     \else
1150       \addcontentsline{toc}{chapter}{#1}%
1151     \fi
1152   \else
1153     \addcontentsline{toc}{chapter}{#1}%
1154   \fi
1155   \chaptermark{#1}%
1156   \addtocontents{lof}{\protect\advspace{10\p@}}%
1157   \addtocontents{lot}{\protect\advspace{10\p@}}%

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1158 \if@twocolumn
1159 \topnewpage[\@makechapterhead{#2}]%
1160 \else
1161 \@makechapterhead{#2}%
1162 \afterheading
1163 \fi}
1164 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1165 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1166 \titleformat{name=\chapter,numberless}[hang]
1167 {\normalfont\Large\bfseries\filcenter}{0pt}{\MakeUppercase}
1168 \titlespacing*{name=\chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
1169 % section: ALL CAPS, normal weight -- the \bfseries here contradicted this very
1170 % comment until v4.1, and made the secondary level as heavy as the primary one,
1171 % defeating the "destaque gradativo" of 2.6. The gradation is now
1172 % chapter = ALL CAPS bold, section = ALL CAPS regular, subsection = Title Case
1173 % bold, subsubsection = Title Case bold italic, as read from the manual.
1174 \titleformat{\section}
1175 {\normalfont\large}{\thesection}{1ex}{\MakeUppercase}
1176 % subsection: bold (author writes Title Case).
1177 \titleformat{\subsection}
1178 {\normalfont\bfseries}{\thesubsection}{1ex}{}
1179 % subsubsection: bold italic.
1180 \titleformat{\subsubsection}
1181 {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{}
1182 % paragraph = quinary section: italic, the author capitalising only the first
1183 % word. 2.6 recommends going as deep as the quinary, and the annotation on the
1184 % 2025 manual (specs/ANOTACOES.md) describes exactly this shape.
1185 %
1186 % book makes \paragraph a RUN-IN heading, so the text continued on the same
1187 % line -- 2.6 requires "o texto deve ser iniciado em outra linha". The positive
1188 % "after" spacing in \titlespacing turns it into a displayed heading.
1189 \titleformat{\paragraph}[hang]
1190 {\normalfont\itshape}{\theparagraph}{1ex}{}
1191 \titlespacing*{\paragraph}{0pt}{2.5ex plus 1ex minus .2ex}{1.5ex plus .2ex}
1192 % book defaults secnumdepth to 2, so \subsubsection and \paragraph came out
1193 % UNNUMBERED however carefully \titleformat named \thesubsubsection. 2.6 has
1194 % the numbering following the whole chain down to the quinary, and the manual's
1195 % own sumario is four levels deep, so tocdepth follows.
1196 \setcounter{secnumdepth}{4}
1197 \setcounter{tocdepth}{4}
1198 % 2.6: "no sumario, as secões devem ser grafadas conforme apresentadas no corpo
1199 % do trabalho". The body sets chapter and section in ALL CAPS (via the
1200 % \MakeUppercase in their \titleformat), but that case change belongs to the
1201 % printed heading alone -- the .toc keeps the title as typed, so the sumario
1202 % came out in ordinary case and contradicted the body.
1203 %
1204 % \l@chapter and \l@section are the two TOC line formatters; wrapping their
1205 % entry argument is enough, and leaves \l@subsection and below untouched, which
1206 % is what we want since those levels are NOT uppercased in the body either.
1207 % The entry carries \numberline{N} plus the title; \MakeUppercase passes the
1208 % digits through unchanged and leaves robust commands (including hyperref's
1209 % link wrappers) intact.
1210 \let\coppe@orig\l@chapter\l@chapter
1211 \renewcommand\l@chapter[2]{\coppe@orig\l@chapter{\MakeUppercase{#1}}{#2}}

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1212 \let\coppe@orig@lsection\l@section
1213 \renewcommand\l@section[2]{\coppe@orig@lsection{\MakeUppercase{#1}}{#2}}
1214 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1215 %      "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1216 \RequirePackage{caption}
1217 % justification=centering makes MULTI-LINE captions centre like the one-line
1218 % ones. Without it, caption's singlelinecheck centres a caption that fits on one
1219 % line and justifies one that wraps, so in a document with mixed caption lengths
1220 % a couple of tables looked left-aligned while the rest looked centred -- which
1221 % is what a recent review noticed about tables 5.1 and 5.2. 2.10 does not
1222 % prescribe the alignment; it does not tolerate two alignments either.
1223 \captionsetup{font=footnotesize,labelsep=endash,justification=centering}
1224 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1225 % caption so it picks up the ABNT captionsetup above.
1226 \RequirePackage{subcaption}
1227 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1228 %      manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1229 %      at the bottom is mandatory). The float package's plaintop style forces the
1230 %      caption above the body for figure, table and the new quadro float. -----
1231 \RequirePackage{float}
1232 \floatstyle{plaintop}
1233 \restylefloat{figure}
1234 \restylefloat{table}
1235 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1236 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1237 % unlike a tabela (rules on top and bottom only).
1238 \providecommand{\quadroname}{\coppestring{frame}}
1239 \providecommand{\listquadroname}{\coppestring{listframe}}
1240 \newfloat{quadro}{htbp}{loq}[chapter]
1241 \floatname{quadro}{\quadroname}
1242 \providecommand{\quadroautorefname}{\quadroname}
1243 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1244 % (float's own \l@quadro lays the number/title out incorrectly).
1245 \let\l@quadro\l@figure
1246 \newcommand\listofquadros{%
1247   \chapter*{\listquadroname}%
1248   \addcontentsline{toc}{chapter}{\listquadroname}%
1249   \mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1250   \@starttoc{loq}}
1251 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1252 % float as a smaller, single-spaced, centred line; it does NOT advance any
1253 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1254 % the label is "Fonte" (pt) / "Source" (en).
1255 \newcommand{\cpsourcename}{\coppestring{source}}
1256 \newcommand{\cpsource}[1]{%
1257   \par\addvspace{\smallskipamount}%
1258   {\footnotesize\singlespacing\centering\cpsourcename: #1\par}}
1259 \let\cpfonte\cpsource
1260 \@ifundefined{source}{\let\source\cpsource}{}%
1261 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1262 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1263 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1264 \newcommand{\newcoppefloat}[3]{%
1265   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%

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1266 \floatname{#1}{#2}%
1267 \expandafter\newcommand\csname listof#1s\endcsname{%
1268 \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%
1269 \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1270 \@starttoc{lo#1}}
1271 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1272 % (filete) from the left margin; footnotesize is the book default. -----
1273 \let\coppe@orig@footnotetext\@footnotetext
1274 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1275 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1276 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1277 % (manual2024 2.6 / proposta P24). -----
1278 \RequirePackage{enumitem}
1279 \newlist{alineas}{enumerate}{2}
1280 \setlist[alineas,1]{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1281 \setlist[alineas,2]{label=--,leftmargin=1.8em,itemsep=0pt}
1282 % --- Page numbering: arabic in the top OUTER corner on textual pages
1283 % (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1284 % \frontmatter (unnumbered by default; roman with the numeraisromanos
1285 % class option). -----
1286 \RequirePackage{fancyhdr}
1287 % \headheight is set to 15pt by geometry up above (see the geometry call),
1288 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1289 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1290 \renewcommand{\headrulewidth}{0pt}}
1291 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1292 % number may leak onto them (e.g. the verso right after the cover).
1293 \renewcommand{\cleardoublepage}{%
1294 \clearpage
1295 \if@twoside
1296 \ifodd\c@page\else
1297 \hbox{}\thispagestyle{empty}\newpage
1298 \if@twocolumn\hbox{}\newpage\fi
1299 \fi
1300 \fi}
1301 \def\CoppeTeX{{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1302 p\kern-.025em e}}\kern-.08em
1303 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}
1304 \newboolean{maledoc}
1305 \setboolean{maledoc}{false}
1306 %
1307 % Class options.
1308 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1309 % fixed language slots:
1310 % * main -- the language of the body, captions, and the main abstract.
1311 % Set by a class option: brazilian (default), english,
1312 % spanish, french, italian, or any user-supplied language
1313 % that ships a coppe-lang-<name>.def file.
1314 % * foreign -- the language used by the foreignabstract environment and
1315 % by \foreign@... registers. Defaults to english (or to
1316 % brazilian when main=english) for back-compat.
1317 % * third -- optional, loaded on demand with \usecoppelanguage{<name>}
1318 % in the preamble (for citations or quotations in a third
1319 % language). Has no class option of its own.

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1320 % Note: the legacy \if@english boolean (defined and toggled here in
1321 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1322 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.lbb)
1323 % tests it. Code that needs to branch on the main language should test
1324 % \ifx\coppe@mainlang\coppe@@englishstr instead.
1325 \def\coppe@mainlang{brazilian}
1326 \def\coppe@foreignlang{english}
1327 \DeclareOption{brazilian}{%
1328   \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1329 \DeclareOption{english}{%
1330   \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1331 \DeclareOption{spanish}{%
1332   \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1333 \DeclareOption{french}{%
1334   \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1335 \DeclareOption{italian}{%
1336   \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1337 \DeclareOption{msc}{%
1338   \newcommand{\@degree}{M.Sc.}
1339   \newcommand{\@degreename}{Mestrado}
1340   \newcommand{\local@degname}{Mestre}
1341   \newcommand{\foreign@degname}{Master}
1342   \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
1343   \newcommand\foreign@doctype{Dissertation}
1344 }
1345 \DeclareOption{dscexam}{%
1346   \newcommand{\@degree}{D.Sc.}
1347   \newcommand{\@degreename}{Doutorado}
1348   \newcommand{\local@degname}{Doutor}
1349   \newcommand{\foreign@degname}{Doctor}
1350   \setboolean{maledoc}{true}
1351   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1352   \newcommand\foreign@doctype{Qualifying Exam}
1353 }
1354 \DeclareOption{mscexam}{%
1355   \newcommand{\@degree}{M.Sc.}
1356   \newcommand{\@degreename}{Mestrado}
1357   \newcommand{\local@degname}{Mestre}
1358   \newcommand{\foreign@degname}{Master}
1359   \setboolean{maledoc}{true}
1360   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1361   \newcommand\foreign@doctype{Qualifying Exam}
1362 }
1363 \DeclareOption{dsc}{%
1364   \newcommand{\@degree}{D.Sc.}
1365   \newcommand{\@degreename}{Doutorado}
1366   \newcommand{\local@degname}{Doutor}
1367   \newcommand{\foreign@degname}{Doctor}
1368   \newcommand\local@doctype{Tese}
1369   \newcommand\foreign@doctype{Thesis}
1370 }
1371 \newif\if@coppenumeric\@coppenumericfalse
1372 \DeclareOption{numbers}{\@coppenumerictrue}
1373 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the

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1374 % pre-textual pages are counted but not numbered (manual2024 2.7).
1375 \newif\if@copperoman\@copperomanfalse
1376 \DeclareOption{numeraisromanos}{\@copperomantrue}

Default line spacing is one-and-a-half; the doublespacing option switches to double spacing. Since setspace is loaded at the end of the class (after hyperref), the actual \onehalfspacing/\doublespacing call is deferred: the option only flips a flag, which the late-load block reads.

1377 \newif\if@coppedoublespace\@coppedoublespacefalse
1378 \DeclareOption{doublespacing}{\@coppedoublespacetrue}
1379 % Two-sided layout is opt-in since v4.1: the 2026 manual dropped the verso
1380 % margins and the deposit is digital (CEPG 246/2023). Given \texttt{twoside},
1381 % the mirrored 3cm inner / 2cm outer layout is restored after \ProcessOptions.
1382 % With no \fichacatalografica file given, the folha adicional leaves a framed
1383 % blank where the ficha goes. The rascunhoficha option fills that blank with the
1384 % class's own LaTeX-composed ficha instead -- useful while writing, never valid
1385 % for deposit: the official ficha comes from the SiBI generator.
1386 % Annex D of the manual draws one signature rule per board member, with the
1387 % identification underneath. The COPPE folha de aprovacao has listed the names
1388 % compactly, without rules, for years. The compact list stays the default and
1389 % the rules are opt-in, so existing theses do not change shape overnight.
1390 % PDF/A output. Opt-in for now: 2.2(d) of the 2026 manual requires the deposited
1391 % file to be PDF/A, but the constraints (embedded fonts, colour profile, no
1392 % encryption, restricted transparency) can trip up a document that pulls in
1393 % arbitrary images or packages, and breaking existing theses to enforce it would
1394 % be worse than letting the author switch it on when depositing.
1395 \newif\if@coppepdfa \@coppepdfafalse
1396 \DeclareOption{pdfa}{\@coppepdfatrue}
1397 \newif\if@coppeassin \@coppeassinfalse
1398 \DeclareOption{assinaturas}{\@coppeassintrue}
1399 \newif\if@copperascunho \@copperascunhofalse
1400 \DeclareOption{rascunhoficha}{\@copperascunhotrue}
1401 \newif\if@coppetwoside \@coppetwosidefalse
1402 \DeclareOption{twoside}{\@coppetwosidetrue}
1403 \DeclareOption{oneside}{\@coppetwosidefalse}

1404 \ProcessOptions\relax
1405 % \LoadClass and geometry ran before \ProcessOptions (see above), so the
1406 % two-sided layout can only be applied here. geometry reads left/right as
1407 % inner/outer once twoside is on, so the 3cm/2cm pair set above becomes the
1408 % mirrored gutter with no further arithmetic. \@twosidetrue is what book's own
1409 % twoside option sets, and is what \cleardoublepage and fancyhdr test.
1410 % In one-sided mode fancyhdr treats every page as odd, so the \fancyhead[RO,LE]
1411 % declarations elsewhere already put the folio in the top right corner of every
1412 % folha, as 2.7 requires.
1413 \if@coppetwoside
1414   \@twosidetrue
1415   \geometry{twoside}%
1416 \fi

1417 % Built-in language names (used by the loader and the dispatcher below).
1418 \def\coppe@@brazilianstr{brazilian}
1419 \def\coppe@@englishstr{english}
1420 % Load Babel with the foreign language first and the main language last
1421 % (Babel treats the LAST option in the list as the main language). With
1422 % pt-main this produces [english,brazilian]{babel} and with en-main

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```

1423 % [brazilian,english] -- identical to the historical \if@english calls.
1424 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1425 \expandafter\RequirePackage\expandafter[\coppe@babelopts]{babel}
1426 % --- Multilingual string dispatcher -----
1427 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1428 % language <lang>. The three retrievers below look strings up either at the
1429 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1430 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1431 \providecommand{\copperdefstring}[3]{%
1432   \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1433 \providecommand{\coppestring}[1]{\@nameuse{coppe@str@\language@ #1}}
1434 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str@\coppe@mainlang @#1}}
1435 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str@\coppe@foreignlang @#1}}
1436 % Load an extra language pack at any point in the preamble (third-language
1437 % slot for citations or quotations beyond the main+foreign pair).
1438 \providecommand{\usecoppelanguage}[1]{%
1439   \def\coppe@@cur{#1}%
1440   \ifx\coppe@@cur\coppe@@brazilianstr\else
1441     \ifx\coppe@@cur\coppe@@englishstr\else
1442       \InputIfFileExists{coppe-lang-#1.def}{}
1443       {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1444     \fi\fi}
1445 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1446 % consulted by the cover/folha renderers once they are migrated in step 2.
1447 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1448 % still drive every visible site, so behavior for pt/en is unchanged.
1449 \providecommand{\titlein}[2]{%
1450   \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1451 % --- Built-in string tables: brazilian + english. The values mirror the
1452 % literals currently scattered through the class; step-1 only populates the
1453 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1454 \copperdefstring{brazilian}{appendix}{Ap\~endice}
1455 \copperdefstring{english}{appendix}{Appendix}
1456 \copperdefstring{brazilian}{annex}{Anexo}
1457 \copperdefstring{english}{annex}{Annex}
1458 \copperdefstring{brazilian}{frame}{Quadro}
1459 \copperdefstring{english}{frame}{Frame}
1460 \copperdefstring{brazilian}{listframe}{Lista de Quadros}
1461 \copperdefstring{english}{listframe}{List of Frames}
1462 \copperdefstring{brazilian}{source}{Fonte}
1463 \copperdefstring{english}{source}{Source}
1464 \copperdefstring{brazilian}{program}{Programa}
1465 \copperdefstring{english}{program}{Program}
1466 \copperdefstring{brazilian}{listprogram}{Lista de Programas}
1467 \copperdefstring{english}{listprogram}{List of Programs}
1468 \copperdefstring{brazilian}{references}{Refer\~encias}
1469 \copperdefstring{english}{references}{References}
1470 \copperdefstring{brazilian}{in:}{em}
1471 \copperdefstring{english}{in:}{in}
1472 \copperdefstring{brazilian}{art}{Ilustra\c{c}\~ao}
1473 \copperdefstring{english}{art}{Art}
1474 \copperdefstring{brazilian}{platform}{Plataforma}
1475 \copperdefstring{english}{platform}{Platform}
1476 \copperdefstring{brazilian}{directedby}{Dire\c{c}\~ao}

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1477 \copperdefstring{english} {directedby} {Directed by}
1478 \copperdefstring{brazilian}{producedby} {Produ\c{c}\~ao}
1479 \copperdefstring{english} {producedby} {Produced by}
1480 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1481 \copperdefstring{english} {listabbreviation}{List of Abbreviations and Acronyms}
1482 \copperdefstring{brazilian}{listsymbol} {Lista de S{\` i}mbolos}
1483 \copperdefstring{english} {listsymbol} {List of Symbols}
1484 \copperdefstring{brazilian}{glossary} {Gloss{\` a}rio}
1485 \copperdefstring{english} {glossary} {Glossary}
1486 \copperdefstring{brazilian}{advisor} {Orientador}
1487 \copperdefstring{english} {advisor} {Advisor}
1488 \copperdefstring{brazilian}{advisors} {Orientadores}
1489 \copperdefstring{english} {advisors} {Advisors}
1490 \copperdefstring{brazilian}{dept} {Programa}
1491 \copperdefstring{english} {dept} {Department}
1492 \copperdefstring{brazilian}{approvedmale} {Aprovado por}
1493 \copperdefstring{brazilian}{approvedfemale}{Aprovada por}
1494 \copperdefstring{brazilian}{approvedonmale} {Aprovado em}
1495 \copperdefstring{brazilian}{approvedonfemale}{Aprovada em}
1496 \copperdefstring{brazilian}{keywordsname} {Palavras-chave}
1497 \copperdefstring{brazilian}{volumename} {Volume}
1498 \copperdefstring{brazilian}{coadvisor} {Coorientador}
1499 \copperdefstring{brazilian}{coadvisors} {Coorientadores}
1500 \copperdefstring{brazilian}{ofname} {de}
1501 \copperdefstring{english} {approvedmale} {Approved by}
1502 \copperdefstring{english} {approvedfemale}{Approved by}
1503 \copperdefstring{english} {approvedonmale} {Approved on}
1504 \copperdefstring{english} {approvedonfemale}{Approved on}
1505 \copperdefstring{english} {keywordsname} {Keywords}
1506 \copperdefstring{english} {volumename} {Volume}
1507 \copperdefstring{english} {coadvisor} {Co-advisor}
1508 \copperdefstring{english} {coadvisors} {Co-advisors}
1509 \copperdefstring{english} {ofname} {of}
1510 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1511 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1512 \copperdefstring{english} {universityname}{Universidade Federal do Rio de Janeiro}
1513 \copperdefstring{brazilian}{cityname} {Rio de Janeiro}
1514 \copperdefstring{english} {cityname} {Rio de Janeiro}
1515 \copperdefstring{brazilian}{statename} {RJ}
1516 \copperdefstring{english} {statename} {RJ}
1517 \copperdefstring{brazilian}{countryname} {Brasil}
1518 \copperdefstring{english} {countryname} {Brasil}
1519 % Month names.
1520 \copperdefstring{brazilian}{monthname1} {Janeiro}
1521 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1522 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1523 \copperdefstring{brazilian}{monthname4} {Abril}
1524 \copperdefstring{brazilian}{monthname5} {Maio}
1525 \copperdefstring{brazilian}{monthname6} {Junho}
1526 \copperdefstring{brazilian}{monthname7} {Julho}
1527 \copperdefstring{brazilian}{monthname8} {Agosto}
1528 \copperdefstring{brazilian}{monthname9} {Setembro}
1529 \copperdefstring{brazilian}{monthname10}{Outubro}
1530 \copperdefstring{brazilian}{monthname11}{Novembro}

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1531 \copperdefstring{brazilian}{monthname12}{Dezembro}
1532 \copperdefstring{english}{monthname1} {January}
1533 \copperdefstring{english}{monthname2} {February}
1534 \copperdefstring{english}{monthname3} {March}
1535 \copperdefstring{english}{monthname4} {April}
1536 \copperdefstring{english}{monthname5} {May}
1537 \copperdefstring{english}{monthname6} {June}
1538 \copperdefstring{english}{monthname7} {July}
1539 \copperdefstring{english}{monthname8} {August}
1540 \copperdefstring{english}{monthname9} {September}
1541 \copperdefstring{english}{monthname10}{October}
1542 \copperdefstring{english}{monthname11}{November}
1543 \copperdefstring{english}{monthname12}{December}
1544 % algorithm2e package option for this language (empty -> english default).
1545 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1546 \copperdefstring{english} {algorithm2eopt}{}
1547 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1548 \copperdefstring{brazilian}{babelname}{brazilian}
1549 \copperdefstring{english} {babelname}{english}
1550 % Auto-load language packs for any non-built-in language listed in babelopts.
1551 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1552 % (defined just above) and need no extra file; others (spanish/french/italian
1553 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
1554 % must already be defined when these run, so the loader sits AFTER the tables.
1555 \@for\coppe@@lang:=\coppe@babelopts\do{%
1556   \edef\coppe@@cur{\coppe@@lang}%
1557   \ifx\coppe@@cur\coppe@@brazilianstr\else
1558     \ifx\coppe@@cur\coppe@@englishstr\else
1559       \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1560       {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1561       {\ClassError{coppe}
1562         {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1563         {Install it next to coppe.cls, or pick a built-in language.}}}%
1564   \fi\fi}
1565 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1566 % Portuguese alias for short, inline quotations (proposta P3).
1567 \RequirePackage{csquotes}
1568 \newcommand{\citacao}[1]{\enquote{#1}}
1569 % --- Code listings & algorithms (TOD03 "tratar listagens"): the listings
1570 % package gives the "Programa" listing (caption on top, ABNT R46) with
1571 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1572 % the language styles below are based on the user's listagens.tex. -----
1573 \RequirePackage{xcolor}
1574 \RequirePackage{listings}
1575 \RequirePackage{fancyvrb}
1576 \DeclareFixedFont{\ttb}{T1}{txxt}{bx}{n}{10}
1577 \DeclareFixedFont{\ttm}{T1}{txxt}{m}{n}{10}
1578 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1579   literate=%
1580   {\~}{\~{a}}1 {\~}{\~{'a}}1 {\~}{\~{a}}1 {\~}{\~{'a}}1 {\~}{\~{'e}}1%
1581   {\~}{\~{'e}}1 {\~}{\~{e}}1 {\~}{\~{'i}}1 {\~}{\~{'i}}1 {\~}{\~{'o}}1%
1582   {\~}{\~{'o}}1 {\~}{\~{o}}1 {\~}{\~{'u}}1 {\~}{\~{'u}}1 {\~}{\~{'c}}1%
1583   {\~}{\~{'c}}1 {\~}{\~{'A}}1 {\~}{\~{'A}}1 {\~}{\~{'A}}1 {\~}{\~{'E}}1%
1584   {\~}{\~{'E}}1 {\~}{\~{'I}}1 {\~}{\~{'I}}1 {\~}{\~{'O}}1 {\~}{\~{'O}}1 {\~}{\~{'O}}1%

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```

1585   {U}{\{\'{U}}}{1%
1586 }
1587 % Babel-aware listing names, set after listings' own babel setup.
1588 \AtBeginDocument{%
1589   \renewcommand\lstlistingname{\coppestring{program}}%
1590   \renewcommand\lstlistlistingname{\coppestring{listprogram}}%
1591 % \listofprogramas in the coppe list style (heading + ToC + running head).
1592 \newcommand\listofprogramas{%
1593   \chapter*{\lstlistlistingname}%
1594   \addcontentsline{toc}{chapter}{\lstlistlistingname}%
1595   \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1596   \@starttoc{lol}}
1597 % Per-language styles + environments (python, java, xml, html, prolog).
1598 % The red hook that marks a wrapped line is built ONCE, into a box, and the
1599 % styles below insert the box. Writing the colour inline -- postbreak=\mbox{%
1600 % \textcolor{red}{\hookrightarrow}\space} -- worked for years and broke the
1601 % moment the pdfa option was switched on: pdfx puts xcolor into a converting
1602 % mode (\selectcolormodel plus \convertcolorsU/D), and a colour command run
1603 % from inside listings' line-breaking machinery then fails with "Missing
1604 % number, treated as zero" at the first wrapped line. Every listing in the
1605 % document dies, so the class's own example could not be compiled as PDF/A at
1606 % all. A box is immune: it is built here, in ordinary horizontal mode where
1607 % xcolor works normally, and listings only ever copies it.
1608 %
1609 % The \sbox waits for \begin{document} because xcolor's colours are not
1610 % defined while the class is still loading.
1611 \newsavebox\coppe@lstarow
1612 \AtBeginDocument{\sbox\coppe@lstarow{\textcolor{red}{\hookrightarrow}\space}}
1613 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1614   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1615   morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1616   emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1617   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1618   extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1619   escapechar=',postbreak=\usebox\coppe@lstarow}}
1620 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
1621   \csname\@lst @SetFirstNumber\endcsname\csname\@lst @SaveFirstNumber\endcsname}
1622 \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!}%
1623 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}}
1624 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1625   numberstyle=\small\ttfamily,numbersep=0.5em,%
1626   morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1627   keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1628   emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
1629   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1630   showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1631   postbreak=\usebox\coppe@lstarow}}
1632 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]{%
1633 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1634   numberstyle=\small\ttfamily,numbersep=0.5em,%
1635   keywordstyle=\ttb\color{rgb}{0,0,0.5},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1636   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1637   extendedchars=true,inputencoding=utf8,breaklines=true,%
1638   postbreak=\usebox\coppe@lstarow}}

```

```

1639 \lstnewenvironment{html}[1][\{\htmlstyle\lstset{#1}\}}{
1640 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
1641   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1642   keywordstyle=\ttb\color{rgb}{0,0,0.5},emph={MyClass,__init__},%
1643   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1644   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1645   breaklines=true,escapechar=',%
1646   postbreak=\usebox\coppe@lstarrow}}
1647 \lstnewenvironment{prolog}[1][\{\prologstyle\lstset{#1}%
1648   \csname\@lst @SetFirstNumber\endcsname\}{\csname\@lst @SaveFirstNumber\endcsname}
1649 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1650   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1651   morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1652   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1653   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1654   breaklines=true,escapechar=',%
1655   postbreak=\usebox\coppe@lstarrow}}
1656 \lstnewenvironment{java}[1][\{\javastyle\lstset{#1}%
1657   \csname\@lst @SetFirstNumber\endcsname\}{\csname\@lst @SaveFirstNumber\endcsname}
1658 \newcommand\javainline[1]{\{\javastyle\lstinline!#1!}}
1659 \newcommand\javaexternal[2][\{\javastyle\lstinputlisting[1]{#2}}
1660 % fancyvrb's Verbatim environment is available for program output; the
1661 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
1662 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
1663 % example,
1664 %   \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
1665 %     ...
1666 %   \end{Verbatim}
1667 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
1668 \renewcommand{\theFancyVerbLine}{\small\ttfamily\arabic{FancyVerbLine}}
1669 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
1670 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
1671 % language (English) -- needed because algorithm2e errors on an empty option.
1672 \edef\coppe@@algotlang{\coppemainstring{algorithm2eopt}}
1673 \ifx\coppe@@algotlang\@empty
1674   \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1675 \else
1676   \expandafter\RequirePackage\expandafter[\coppe@@algotlang,linesnumbered,lined,algochapter,r
1677 \fi
1678 % algorithm2e composes its own caption and therefore ignores the class-wide
1679 % \captionsetup{font=footnotesize,labelsep=endash}. Left alone it printed
1680 % "Algoritmo 6.1:Titulo" in bold at body size, while every other illustration
1681 % printed "Figura 6.1 -- Titulo" in footnotesize -- 2.10 asks for the
1682 % designating word, the number, a TRAVESSAO and the title, uniformly.
1683 %
1684 % The separator and the caption fonts are therefore set to match. The 'ruled'
1685 % layout (rules above and below) is kept: 2.10 prescribes where the caption and
1686 % the source go, not whether the float is delimited, and the rules read here
1687 % like the top-and-bottom delimitation a table carries.
1688 \SetAlgoCaptionSeparator{\textendash}
1689 \SetAlCapFnt{\footnotesize}
1690 \SetAlCapNameFnt{\footnotesize}
1691 \SetAlCapSty{textnormal}
1692 \SetAlCapNameSty{textnormal}

```

```

1693 \SetKwFor{Para}{para}{faça}{fim do para}
1694 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1695 % setspace + hyperref + biblatex are loaded at the very end of the class
1696 % (see the block just before %</class>), so their hooks see every macro the
1697 % class defines, and the load order setspace -> hyperref -> biblatex is
1698 % respected.

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

1699 \newcommand\coppe@deptcode{}
1700 \newcommand\meta@deptname{}
1701 \newcommand\department[1]{%
1702 % The programme CODE is kept alongside the resolved name, and each name is
1703 % declared TWICE. \local@deptname keeps the TeX accent escapes with braces
1704 % (Computa{\c c}{\~ a}o) that the cover and the folha de rosto typeset;
1705 % \meta@deptname is the same name in literal UTF-8, and is the one that goes
1706 % into the XMP. pdfx copies whatever it is given straight through, so feeding it
1707 % the escaped form produced "Computa{c}o" in dc:description -- which is why the
1708 % subject line used to fall back to the bare programme code. Both forms are
1709 % written out here rather than derived from one another because there is no
1710 % reliable way to strip TeX accent syntax at the point where the .xmpdata is
1711 % written, and a table of thirteen entries is cheaper than that machinery.
1712 \gdef\coppe@deptcode{#1}%
1713 \ifthenelse{\equal{#1}{PEB}}{
1714   {\global\def\local@deptname{Engenharia Biom{\` e}dica}
1715    \global\def\meta@deptname{Engenharia Biomédica}
1716    \global\def\foreign@deptname{Biomedical Engineering}}{}
1717 \ifthenelse{\equal{#1}{PEC}}{
1718   {\global\def\local@deptname{Engenharia Civil}
1719    \global\def\meta@deptname{Engenharia Civil}
1720    \global\def\foreign@deptname{Civil Engineering}}{}
1721 \ifthenelse{\equal{#1}{PEE}}{
1722   {\global\def\local@deptname{Engenharia El{\` e}trica}
1723    \global\def\meta@deptname{Engenharia Elétrica}
1724    \global\def\foreign@deptname{Electrical Engineering}}{}
1725 \ifthenelse{\equal{#1}{PEM}}{
1726   {\global\def\local@deptname{Engenharia Mec{\` a}nica}
1727    \global\def\meta@deptname{Engenharia Mecânica}
1728    \global\def\foreign@deptname{Mechanical Engineering}}{}
1729 \ifthenelse{\equal{#1}{PEMM}}{
1730   {\global\def\local@deptname{Engenharia Metal{\` u}rgica e de Materiais}
1731    \global\def\meta@deptname{Engenharia Metalúrgica e de Materiais}
1732 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
1733 \ifthenelse{\equal{#1}{PEN}}{
1734   {\global\def\local@deptname{Engenharia Nuclear}
1735    \global\def\meta@deptname{Engenharia Nuclear}
1736    \global\def\foreign@deptname{Nuclear Engineering}}{}
1737 \ifthenelse{\equal{#1}{PEN0}}{
1738   {\global\def\local@deptname{Engenharia Oce{\` a}nica}
1739    \global\def\meta@deptname{Engenharia Oceânica}
1740    \global\def\foreign@deptname{Ocean Engineering}}{}
1741 \ifthenelse{\equal{#1}{PPE}}{
1742   {\global\def\local@deptname{Planejamento Energ{\` e}tico}

```

```

1743 \global\def\meta@deptname{Planejamento Energético}
1744 \global\def\foreign@deptname{Energy Planning}}{}
1745 \ifthenelse{\equal{#1}{PEP}}{
1746   {\global\def\local@deptname{Engenharia de Produ{\c c}{\~ a}o}
1747   \global\def\meta@deptname{Engenharia de Produção}
1748   \global\def\foreign@deptname{Production Engineering}}{}
1749 \ifthenelse{\equal{#1}{PEQ}}{
1750   {\global\def\local@deptname{Engenharia Qu{\` i}mica}
1751   \global\def\meta@deptname{Engenharia Química}
1752   \global\def\foreign@deptname{Chemical Engineering}}{}
1753 \ifthenelse{\equal{#1}{PESC}}{
1754   {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\~ a}o}
1755   \global\def\meta@deptname{Engenharia de Sistemas e Computação}
1756   \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1757 \ifthenelse{\equal{#1}{PET}}{
1758   {\global\def\local@deptname{Engenharia de Transportes}
1759   \global\def\meta@deptname{Engenharia de Transportes}
1760   \global\def\foreign@deptname{Transportation Engineering}}{}
1761 \ifthenelse{\equal{#1}{PENT}}{
1762   {\global\def\local@deptname{Engenharia de Nanotecnologia}
1763   \global\def\meta@deptname{Engenharia de Nanotecnologia}
1764   \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1765 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

1766 \renewcommand\title[1]{%
1767   \global\def\local@title{#1}%
1768   \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
1769 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

1770 \newcommand\foreigntitle[1]{%
1771   \global\def\foreign@title{#1}%
1772   \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
1773 }

```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```

1774 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
1775 % Subtitle, mirroring the title machinery slot for slot: \subtitle feeds the
1776 % brazilian slot, \foreignsubtitle the english one, \subtitlein any other.
1777 % 3.1.1 and 3.1.2.1.1(c) require the subtitle on the capa and on the folha de
1778 % rosto, subordinated to the title by a colon.
1779 \newcommand\subtitle[1]{%
1780   \expandafter\gdef\csname coppe@subtitle@brazilian\endcsname{#1}}
1781 \newcommand\foreignsubtitle[1]{%
1782   \expandafter\gdef\csname coppe@subtitle@english\endcsname{#1}}
1783 \newcommand\subtitlein[2]{%
1784   \expandafter\gdef\csname coppe@subtitle@#1\endcsname{#2}}

```



```

1785 \newcommand\coppe@selectsubtitle{\@nameuse{coppe@subtitle@\coppe@mainlang}}
1786 % Title as the capa and the folha de rosto print it. Annexes A and B show the
1787 % title in caps and the subtitle in ordinary case after a colon, so the case
1788 % change is applied to the title alone.
1789 \newcommand\coppe@titleblock{%
1790   \MakeUppercase{\coppe@selecttitle}%
1791   \@ifundefined{coppe@subtitle@\coppe@mainlang}{\coppe@selectsubtitle}}
1792 % Number of volumes (3.1.1 and 3.1.2.1.1(d)): printed only when there is more
1793 % than one, since a single-volume work says nothing about volumes.
1794 % Coorientador (3.1.2.1.1(f)). Same shape as \advisor, including the optional
1795 % institution; the label switches to the plural on the second call, as the
1796 % advisor label already did.
1797 \newcount\@coadvisor\@coadvisor0
1798 \newcommand\coadvisor[5][\%
1799   \global\@namedef{CoppeCoadvisorInst:\expandafter\the\@coadvisor}{#1}
1800   \global\@namedef{CoppeCoadvisorTitle:\expandafter\the\@coadvisor}{#2}
1801   \global\@namedef{CoppeCoadvisorName:\expandafter\the\@coadvisor}{#3}
1802   \global\@namedef{CoppeCoadvisorSurname:\expandafter\the\@coadvisor}{#4}
1803   \global\@namedef{CoppeCoadvisorDegree:\expandafter\the\@coadvisor}{#5}
1804   \global\advance\@coadvisor by 1
1805 }
1806 \newcommand\local@coadvisorstring{%
1807   \ifnum\@coadvisor>1
1808     \coppemainstring{coadvisors}%
1809   \else
1810     \coppemainstring{coadvisor}%
1811   \fi}
1812 \newcommand\coppe@nvolumes{1}
1813 \newcommand\coppe@thisvolume{1}
1814 \newcommand\volumes[1]{\gdef\coppe@nvolumes{#1}}
1815 \newcommand\volume[1]{\gdef\coppe@thisvolume{#1}}
1816 \newcommand\coppe@volumeline{%
1817   \ifnum\coppe@nvolumes>1
1818     \par\vspace*{4mm}%
1819     \coppemainstring{volumename}\space\coppe@thisvolume\space
1820     \coppemainstring{ofname}\space\coppe@nvolumes
1821 % The closing \par matters: on the capa the next thing is \vspace{\stretch{3}},
1822 % which does not end a paragraph, so without it the city ran on into the volume
1823 % line ("Volume 2 de 3 RIO DE JANEIRO").
1824   \par
1825   \fi}

```

\advisor Defines globally the title, name and academic degree of the advisors.

```

1826 \newcount\@advisor\@advisor0
1827 % The optional first argument is the member's institution, required by
1828 % 3.1.2.1.3(e). Omitting it keeps every pre-4.1 call working unchanged.
1829 \newcommand\advisor[5][\%
1830   \global\@namedef{CoppeAdvisorInst:\expandafter\the\@advisor}{#1}
1831   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#2}
1832   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#3}
1833   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#4}
1834   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#5}
1835   \global\advance\@advisor by 1
1836   \ifnum\@advisor>1

```

```

1837 \renewcommand\local@advisorstring{Orientadores}
1838 \renewcommand\foreign@advisorstring{Advisors}
1839 \fi
1840 }

```

\examiner

```

1841 \newcount\@examiner\@examiner0
1842 % Until v4.1 this stored only "#1 #2" and DISCARDED the third argument: the
1843 % titulacao every document passed was silently thrown away and never printed,
1844 % though 3.1.2.1.3(e) requires it. Each part is now kept separately, and the
1845 % optional first argument carries the institution.
1846 % CoppeExaminer: is retained with its old meaning for anything reading it.
1847 \newcommand\examiner[4][\%
1848 \global\@namedef{CoppeExaminerInst:\expandafter\the\@examiner}{#1}
1849 \global\@namedef{CoppeExaminerTitle:\expandafter\the\@examiner}{#2}
1850 \global\@namedef{CoppeExaminerName:\expandafter\the\@examiner}{#3}
1851 \global\@namedef{CoppeExaminerDegree:\expandafter\the\@examiner}{#4}
1852 \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#2\ #3}
1853 \global\advance\@examiner by 1
1854 }

```

\author It was redefined to allow the identification of the author's first names and surname.

```

1855 \renewcommand\author[2]{%
1856 \global\def\@authname{#1}
1857 \global\def\@authsurn{#2}
1858 }

```

\date Sets the *document's* nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (\coppe@docmonth / \coppe@docyear) rather than mutating TeX's \month / \year primitives, which would clobber \today.

```

1859 \renewcommand\date[2]{%
1860 \def\coppe@docmonth{#1}%
1861 \def\coppe@docyear{#2}}

```

\local@monthname

```

1862 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
1863 Janeiro\or Fevereiro\or Mar{\c c}o\or Abril\or Maio\or Junho\or
1864 Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}

```

\foreign@monthname

```

1865 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
1866 January\or February\or March\or April\or May\or June\or
1867 July\or August\or September\or October\or November\or December\fi}

```

\keyword

```

1868 \newcounter{keywords}
1869 \newcommand\keyword[1]{%
1870 \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
1871 \global\addtocounter{keywords}{1}
1872 }
1873 % Keywords for the foreign abstract. Annex F of the manual shows the English

```

```

1874 % abstract closing with English keywords, and \keyword alone cannot serve both
1875 % pages. When no \foreignkeyword is given the foreign abstract falls back to
1876 % the \keyword list, which is what a document written before v4.1 expects.
1877 \newcounter{foreignkeywords}
1878 \newcommand\foreignkeyword[1]{%
1879   \global\@namedef{CoppeForeignKeyword:\expandafter\the\c@foreignkeywords}{#1}
1880   \global\addtocounter{foreignkeywords}{1}
1881 }
1882 % Keywords for the optional third abstract (Portuguese, for the banca). Same
1883 % shape and same fallback as \foreignkeyword: without it the Portuguese page
1884 % would close with the main-language keywords under a Portuguese label, which
1885 % is what a Spanish-, French- or Italian-main work would produce.
1886 \newcounter{braziliankeywords}
1887 \newcommand\braziliankeyword[1]{%
1888   \global\@namedef{CoppeBrazilianKeyword:\expandafter\the\c@braziliankeywords}{#1}
1889   \global\addtocounter{braziliankeywords}{1}
1890 }
1891 % 3.1.2.1.4: the keywords close the resumo, preceded by the term, a colon, and
1892 % separated from one another by semicolons.
1893 % #1 label, #2 csname prefix, #3 how many.
1894 \newcommand\coppe@printkeywords[3]{%
1895   \ifnum#3>0
1896     \par\vspace*{7mm}%
1897     \noindent{\singlespacing\textbf{#1:}\enspace
1898       \count1=0
1899       \@whilenum\count1<#3\do{%
1900         \ifnum\count1>0 ;\space\fi
1901         \csname #2\the\count1\endcsname
1902         \advance\count1 by 1}%
1903       .\par}%
1904   \fi}

```

\areaconcentracao Data for the folha adicional (Coleta CAPES) and for the natureza block of the
\linhapesquisa folha de rosto and folha de aprovacao. Every one of them defaults to empty;
\tipoproducao \makefolhaadicional prints an empty rule where a value is missing, so the pro-
\projetovinculado gramme can complete the sheet by hand if need be.

```

\nomeprojeto 1905 \newcommand\coppe@areaconc{}
\agenciafomento 1906 \newcommand\areaconcentracao[1]{\gdef\coppe@areaconc{#1}}
1907 \newcommand\coppe@dataaprov{}
1908 \newcommand\dataaprovacao[1]{\gdef\coppe@dataaprov{#1}}
1909 \newcommand\coppe@linhapesq{}
1910 \newcommand\linhapesquisa[1]{\gdef\coppe@linhapesq{#1}}
1911 \newcommand\coppe@tipoproducao{}
1912 \newcommand\tipoproducao[1]{\gdef\coppe@tipoproducao{#1}}
1913 \newcommand\coppe@projetovinc{}
1914 \newcommand\projetovinculado[1]{\gdef\coppe@projetovinc{#1}}
1915 \newcommand\coppe@nomeprojeto{}
1916 \newcommand\nomeprojeto[1]{\gdef\coppe@nomeprojeto{#1}}
1917 \newcounter{coppeagencias}
1918 \newcommand\agenciafomento[2]{%
1919   \global\@namedef{CoppeAgencia:\expandafter\the\c@coppeagencias}{#1}%
1920   \global\@namedef{CoppeAgenciaSigla:\expandafter\the\c@coppeagencias}{#2}%
1921   \global\addtocounter{coppeagencias}{1}}

```

`\freeconfig` This command allows easy changing of core class parameters.

```

1922 \newcommand\freeconfig[6]{%
1923 \providecommand\@degree{}
1924 \renewcommand\@degree{#1}
1925 \providecommand\@degreename{}
1926 \renewcommand\@degreename{#2}
1927 \providecommand\local@degname{}
1928 \renewcommand\local@degname{#3}
1929 \providecommand\foreign@degname{}
1930 \renewcommand\foreign@degname{#4}
1931 \providecommand\local@doctype{}
1932 \renewcommand\local@doctype{#5}
1933 \providecommand\foreign@doctype{}
1934 \renewcommand\foreign@doctype{#6}%
1935 }
1936 % \end{macrocode}
1937 % \end{macro}
1938 %
1939 % \begin{macro}{\frontmatter}
1940 % The number of pages for both frontmatter and mainmatter printed
1941 % in the cataloging details page is computed by means of simple
1942 % LaTeX labels.
1943 % \begin{macrocode}
1944 \renewcommand\frontmatter{%
1945 \cleardoublepage
1946 \@mainmatterfalse
1947 % The count already started at the folha de rosto (see \maketitle) and simply
1948 % continues here: 2.7 requires every folha from the folha de rosto onwards to
1949 % be counted sequentially, printing arabic numerals only from the first textual
1950 % folha. Resetting the counter here -- as this class did until v4.1 -- made the
1951 % Introducao come out as page 1, which the norm does not allow.
1952 %
1953 % The numeraisromanos option keeps the legacy behaviour (lowercase roman in the
1954 % pre-textual part, arabic restarting at 1 in the textual part). That departs
1955 % from 2.7, which has the pre-textual folhas counted but NOT numbered, so the
1956 % option warns.
1957 \if@copperoman
1958 \ClassWarningNoLine{coppe}{Option ‘numeraisromanos’ prints roman numerals
1959 on the pre-textual folhas and restarts the arabic count at 1 in the
1960 textual part. Section 2.7 of the SiBI manual (9th rev. ed., 2026) has
1961 those folhas counted but not numbered, and the arabic numbering
1962 continuing the count. Drop the option to comply}%
1963 \pagenumbering{roman}%
1964 \fi
1965 \thispagestyle{empty}
1966 \makefrontpage
1967 \clearpage
1968 \if@copperoman
1969 \fancypagestyle{plain}{\fancyhf{} \fancyhead[R0,LE]{\thepage}}%
1970 \renewcommand{\headrulewidth}{0pt}}%
1971 \pagestyle{coppe}%
1972 \else
1973 \fancypagestyle{plain}{\fancyhf{} \renewcommand{\headrulewidth}{0pt}}%
1974 \pagestyle{empty}%

```

```

1975 \fi
1976 }

```

\mainmatter

```

1977 \renewcommand\mainmatter{%
1978 \coppe@mainBegin
1979 \cleardoublepage
1980 \@mainmattertrue
1981 \fancypagestyle{plain}{\fancyhf{}\fancyhead[RO,LE]{\thepage}%
1982 \renewcommand{\headrulewidth}{0pt}}%
1983 \pagestyle{coppe}%
1984 % No \pagenumbering{arabic} in the default path: it would reset the counter to
1985 % 1 and make the Introducao folha 1, contrary to 2.7. The counter has been
1986 % arabic and running since the folha de rosto; all that changes here is that
1987 % the folio starts being PRINTED. Only the legacy numeraisromanos path, which
1988 % switched to roman in the front matter, has to switch back -- and there the
1989 % restart at 1 is intrinsic to that (non-conforming) scheme.
1990 \if@copperoman\pagenumbering{arabic}\fi

```

\backmatter

```

1991 \renewcommand\backmatter{%
1992 \if@openright
1993 \cleardoublepage
1994 \else
1995 \clearpage
1996 \fi}
1997 %

```

\coppetexfinalpage Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the L^AT_EX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with `\renewcommand` if the document uses a non-default font or follows an earlier edition of the norm.

```

1998 \providecommand{\coppefinalengine}{%
1999 \ifPDFTeX\hologo{pdfLaTeX}\fi
2000 \ifLuaTeX\hologo{LuaLaTeX}\fi
2001 \ifXeTeX\hologo{XeLaTeX}\fi}
2002 \providecommand{\coppefinalfont}{Computer Modern (padr~ao \LaTeX{})}
2003 \providecommand{\coppefinalmanual}{%
2004 \emph{Norma para a Elaborac~ao Gr~afica de Teses e Dissertac~oes
2005 da COPPE/UFRJ}}
2006 \newcommand{\coppetexfinalpage}{%
2007 \clearpage
2008 % Force the colophon onto a verso (even) page so it faces the rear cover.
2009 % If the current page is odd (recto), fill it with an empty verso first.
2010 \if@twoside

```

```

2011 \ifodd\c@page
2012 \hbox{}\thispagestyle{empty}\newpage
2013 \fi
2014 \fi
2015 \thispagestyle{empty}%
2016 \null\vfill
2017 \begin{otherlanguage}{brazilian}%
2018 \begin{center}
2019 {\footnotesize\setstretch{1.0}}%
2020 \textsc{Colof\~ao}\par
2021 \vspace{0.7em}
2022 \begin{minipage}{0.75\textwidth}\centering
2023 Este documento foi composto em \LaTeX{} com \coppefinalengine{}
2024 em \today, na fonte \coppefinalfont. A bibliografia foi processada
2025 por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
2026 vers\~ao\~\coppe@version, dispon\~\i vel no reposit\~\orio GitHub
2027 \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
2028 conformidade com a \coppefinalmanual.
2029 \end{minipage}\par
2030 \end{center}
2031 \end{otherlanguage}%
2032 \vspace*{2cm}%
2033 \clearpage}

```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```

2034 \newcommand\makecover{%
2035 \begin{titlepage}%
2036 \centering
2037 \makebox[\linewidth]{%
2038 \includegraphics[height=2cm]{ufrj-logo}\hfill
2039 \includegraphics[height=2cm]{coppe-logo}}\par
2040 \vspace{1.5cm}
2041 {\singlespacing
2042 \MakeUppercase{\local@universityname}\par
2043 \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de \\\
2044 P{\~\o}s-Gradua{\c c}{\~\a}o e Pesquisa de Engenharia}}\par
2045 \MakeUppercase{Programa de \local@deptname}\par}
2046 \vspace{\stretch{2}}
2047 \textbf{\MakeUppercase{\textbf{\@authname\ \@authsur}}}\par
2048 \vspace{\stretch{2}}
2049 \nohyphens{\coppe@titleblock}\par
2050 \coppe@volumeline
2051 \vspace{\stretch{3}}
2052 \MakeUppercase{\local@cityname}\par
2053 \coppe@docyear
2054 \end{titlepage}%
2055 }

```

\maketitle

```
2056 \renewcommand\maketitle{%
2057 % The capa is neither counted nor numbered (2.7), so it lives in its own
2058 % throwaway numbering scheme; \pagenumbering{arabic} right after it restarts
2059 % the counter, making the FOLHA DE ROSTO page 1 -- which is where 2.7 says the
2060 % count begins. Nothing is printed here: \frontmatter keeps \pagestyle{empty}
2061 % until the textual part starts.
2062 \pagenumbering{alph}
2063 \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
2064 \makecover
2065 \pagenumbering{arabic}
2066 \begin{titlepage}
2067 \begin{flushleft}
2068 \vspace*{1.5mm}
2069 \setlength\baselineskip{0pt}
2070 \setlength\parskip{1mm}
2071 \makebox[\linewidth]{%
2072 \includegraphics[height=2cm]{ufrj-logo}\hfill
2073 \includegraphics[height=2cm]{coppe-logo}}
2074 \end{flushleft}
2075 \vspace{1.05cm}
2076 \begin{center}
2077 \nohyphens{\@authname\ \@authsurv}\par
2078 \vspace*{3cm}
2079 \nohyphens{\coppe@titleblock}\par
2080 \coppe@volumeline
2081 \end{center}
2082 \vspace*{2.1cm}
2083 \begin{flushright}
2084 % 2.4: on the folha de rosto the natureza/objetivo/instituicao block is set
2085 % SINGLE spaced, and aligned from the middle of the mancha grafica to the right
2086 % margin. 0.5\textwidth tracks the mancha, so the block follows automatically if
2087 % the margins ever change -- the old fixed 8.45cm did not.
2088 \begin{minipage}{0.5\textwidth}
2089 \singlespacing\frontcover@maintext
2090 \end{minipage}\par
2091 \end{flushright}
2092 % Only the natureza block is indented to the right (2.4). The linha de pesquisa
2093 % and the orientadores return to the LEFT MARGIN, as the CAPES model of the
2094 % folha de rosto and Annex B of the manual show. Until v4.1 they sat inside the
2095 % right-hand block, which is what a recent review flagged.
2096 \vspace*{12mm}
2097 {\singlespacing
2098 \ifthenelse{\equal{\coppe@linhapesq}{}}{\}%
2099 \noindent Linha de pesquisa: \coppe@linhapesq\par
2100 \vspace*{5mm}}%
2101 \nohyphens{%
2102 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}}
2103 \local@advisorstring: &
2104 \count1=0
2105 \toks@={}
2106 \@whilenum \count1<\@advisor \do{%
2107 \ifcase\count1 % same as \ifnum0=\count1
2108 \toks@=\expandafter{\curname CoppeAdvisorName:\the\count1%
```

```

2109         \expandafter\endcsname\expandafter\space%
2110         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2111     \else
2112         \toks@=\expandafter\expandafter\expandafter{%
2113             \expandafter\the\expandafter\toks@%
2114             \expandafter&\expandafter\space%
2115             \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2116             \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2117         }%
2118     \fi
2119     \advance\count1 by 1}
2120     \the\toks@
2121 \end{tabularx}}}%
2122 \ifnum\@coadvisor>0
2123     \par\vspace*{2mm}%
2124     \nohyphens{%
2125     \noindent\begin{tabularx}{\textwidth}[b]{\@{1}{ }>{\raggedright\arraybackslash}X@{}}
2126         \local@coadvisorstring: &
2127         \count1=0
2128         \toks@={}
2129         \@whilenum \count1<\@coadvisor \do{%
2130             \ifcase\count1
2131                 \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2132                     \expandafter\endcsname\expandafter\space%
2133                     \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2134             \else
2135                 \toks@=\expandafter\expandafter\expandafter{%
2136                     \expandafter\the\expandafter\toks@%
2137                     \expandafter&\expandafter\space%
2138                     \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2139                     \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\
2140                 }%
2141             \fi
2142             \advance\count1 by 1}
2143         \the\toks@
2144     \end{tabularx}}}%
2145 \fi
2146 \par}
2147 \vspace*{\fill}
2148 % 3.1.2.1.1(g): the folha de rosto carries the ANO DE DEPOSITO alone -- the year
2149 % the revised final version is handed to the Programa. The month was never asked
2150 % for here (the capa, Annex A, also shows only the year), and printing it
2151 % invited confusion with the defence date. \date still takes month and year:
2152 % the month is what the abstract pages and the colophon use.
2153 \begin{center}
2154 \local@cityname\par
2155 \coppe@docyear
2156 \end{center}
2157 \end{titlepage}
2158 % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
2159 % the title page is a recto, so this lands on the immediately following verso,
2160 % ahead of the approval sheet produced by \frontmatter.
2161 % LaTeX's titlepage environment ends with \setcounter{page}{1} whenever the
2162 % document is one-sided (see \endtitlepage in classes.dtx). That silently ate

```



```

2163 % the folha de rosto from the count, putting the folha de aprovacao at 1 instead
2164 % of 2. The count is therefore restated here: the folha de rosto WAS folha 1, so
2165 % whatever follows it is folha 2.
2166 \setcounter{page}{2}%
2167 % The page carrying the ficha catalografica is explicitly excluded from the
2168 % count -- 2.7 of the 2026 manual ("a folha adicional que contem a ficha
2169 % catalografica nao pode ser contada ou numerada"), and the 2025 edition said
2170 % the same of the verso of the folha de rosto. Hence the counter is stepped
2171 % back after it.
2172 \ifthenelse{\boolean{maledoc}}{}{%
2173 \makefolhaadicional\clearpage\addtocounter{page}{-1}}%
2174 \global\let\maketitle\relax%
2175 \global\let\and\relax}

2176 % 3.1.2.1.3 lists the elements of the folha de aprovacao in order: (a) the
2177 % AUTHOR, (b) the title in full, (c) the natureza/objetivo/instituicao/area de
2178 % concentracao, (d) the date of approval, (e) the examining board. Until v4.1
2179 % this page opened with the title in caps and the author under it, and set the
2180 % natureza as a wall of uppercase (\frontpage@maintext). Annex D shows neither.
2181 % One board member. \coppe@membroid is the identification line required by
2182 % 3.1.2.1.3(e) -- name, titulacao and institution -- with the last two omitted
2183 % when not given, so a document that never supplied them looks as it did before.
2184 \newcommand\coppe@membroid[4]{%
2185 #1\ #2%
2186 \ifthenelse{\equal{#3}{} }{, #3}%
2187 \ifthenelse{\equal{#4}{} }{, #4}}
2188 % The gap above each signature rule is \coppe@membrogap, set from the size of
2189 % the board just before the list is assembled (see \makefrontpage). A fixed 9mm
2190 % fitted five members and overflowed at seven -- the sheet silently continued
2191 % onto a second page, which also printed a folio in the pre-textual part, where
2192 % 2.7 forbids one. The shrink lives in the length itself -- \newlength makes a
2193 % skip register, so "9mm minus 2mm" survives \setlength and reaches \vspace as
2194 % real glue, and a title long enough to wrap an extra line is absorbed instead
2195 % of costing a page. It has to go through \vspace and not a bare \vskip: the
2196 % board list is emitted inside \nohyphens, i.e. in horizontal mode, where
2197 % \vskip is an error and TeX then typesets the "plus ... minus ..." after it as
2198 % ordinary text -- which is exactly what landed on the page the first time.
2199 \newlength\coppe@membrogap
2200 \setlength\coppe@membrogap{9mm minus 2mm}
2201 \newcount\coppe@bancasize
2202 \newcommand\coppe@membro[4]{%
2203 \if@coppeassin
2204 \vspace{\coppe@membrogap}%
2205 \begin{center}
2206 \rule{95mm}{0.4pt}\par
2207 \nopagebreak{\footnotesize\coppe@membroid{#1}{#2}{#3}{#4}}\par
2208 \end{center}%
2209 \else
2210 \noindent\hspace*{8mm}%
2211 \begin{minipage}{\dimexpr\textwidth-8mm\relax}
2212 \coppe@membroid{#1}{#2}{#3}{#4}
2213 \end{minipage}\par\vspace{2mm}%
2214 \fi}
2215 \newcommand\makefrontpage{%

```

```

2216 \begin{center}
2217   \nohyphens{\@authname\ \@authsur}\par
2218   \vspace*{7mm}
2219   \sloppy\nohyphens{\coppe@selecttitle
2220     \@ifundefined{coppe@subtitle@\coppe@mainlang}{\{:
2221       \coppe@selectsubtitle}}\par
2222 \end{center}\par
2223 \vspace*{8mm}
2224 % Same block, same indentation and same single spacing as the folha de rosto:
2225 % 2.4 names the two pages together.
2226 \begin{flushright}
2227 \begin{minipage}{0.5\textwidth}
2228 \single-spacing\frontcover@maintext
2229 \end{minipage}
2230 \end{flushright}
2231 \vspace*{12mm}
2232 % (d) date of approval. Annex D prints "Aprovada em:" followed by a blank to be
2233 % filled in; \dataaprovacao supplies it when the date is already known.
2234 \noindent\local@approvedonname:\enspace
2235 \ifthenelse{\equal{\coppe@dataaprov}{}}{\hrulefill}{\coppe@dataaprov}\par
2236 \vspace*{12mm}
2237 % (e) the examining board, with the ORIENTADOR FIRST, as president of the board
2238 % (3.1.2.1.3(e)). Until v4.1 the advisors were printed in a block of their own
2239 % above a second list headed "Aprovada por:", which split the board in two and
2240 % did not mark the advisor as its president. They are now one list.
2241 %
2242 % The list is assembled into \coppe@bancalist with \g@addto@macro instead of
2243 % being built inside a tabularx: each entry now carries four fields (treatment,
2244 % name, titulacao, institution) and the old \toks@/\expandafter accumulation
2245 % could not carry them without becoming unreadable. \protected@edef keeps
2246 % accents and other protected commands in the names intact.
2247 % Size of the board decides the spacing above. The thresholds are measured, not
2248 % guessed: with the 9mm gap a seven-member board pushed three members onto a
2249 % second page, 7mm brought six back onto one sheet and 5mm did the same for
2250 % seven and eight. Boards of five or fewer keep the original 9mm, so every
2251 % document written before this change renders exactly as it did.
2252 \coppe@bancasize=\@advisor
2253 \advance\coppe@bancasize by \@coadvisor
2254 \advance\coppe@bancasize by \@examiner
2255 \ifnum\coppe@bancasize>6
2256   \setlength\coppe@membrogap{5mm minus 2mm}%
2257 \else\ifnum\coppe@bancasize>5
2258   \setlength\coppe@membrogap{7mm minus 2mm}%
2259 \fi\fi
2260 \gdef\coppe@bancalist{}%
2261 \count1=0
2262 \@whilenum\count1<\@advisor\do{%
2263   \protected@edef\coppe@tmp{\noexpand\coppe@membro
2264     {\csname CoppeAdvisorTitle:\the\count1\endcsname}%
2265     {\csname CoppeAdvisorName:\the\count1\endcsname\space
2266       \csname CoppeAdvisorSurname:\the\count1\endcsname}%
2267     {\csname CoppeAdvisorDegree:\the\count1\endcsname}%
2268     {\csname CoppeAdvisorInst:\the\count1\endcsname}}%
2269   \expandafter\g@addto@macro\expandafter\coppe@bancalist

```

```

2270     \expandafter{\coppe@tmp}%
2271     \advance\count1 by 1}%
2272 % Coorientadores come right after the orientadores and before the rest of the
2273 % board, keeping the order 3.1.2.1.3(e) prescribes.
2274 \count1=0
2275 \@whilenum\count1<\@coadvisor\do{%
2276     \protected@edef\coppe@tmp{\noexpand\coppe@membro
2277         {\csname CoppeCoadvisorTitle:\the\count1\endcsname}%
2278         {\csname CoppeCoadvisorName:\the\count1\endcsname\space
2279         \csname CoppeCoadvisorSurname:\the\count1\endcsname}%
2280         {\csname CoppeCoadvisorDegree:\the\count1\endcsname}%
2281         {\csname CoppeCoadvisorInst:\the\count1\endcsname}}%
2282     \expandafter\g@addto@macro\expandafter\coppe@bancalist
2283     \expandafter{\coppe@tmp}%
2284     \advance\count1 by 1}%
2285 \count1=0
2286 \@whilenum\count1<\@examiner\do{%
2287     \protected@edef\coppe@tmp{\noexpand\coppe@membro
2288         {\csname CoppeExaminerTitle:\the\count1\endcsname}%
2289         {\csname CoppeExaminerName:\the\count1\endcsname}%
2290         {\csname CoppeExaminerDegree:\the\count1\endcsname}%
2291         {\csname CoppeExaminerInst:\the\count1\endcsname}}%
2292     \expandafter\g@addto@macro\expandafter\coppe@bancalist
2293     \expandafter{\coppe@tmp}%
2294     \advance\count1 by 1}%
2295 \noindent\local@approvedname:\par
2296 \nohyphens{\singlespacing\coppe@bancalist}\par
2297 % No city/date block at the foot: 3.1.2.1.3 ends the folha de aprovacao at the
2298 % examining board, and Annex D shows nothing below the signature lines. The date
2299 % that belongs on this page is the date of APPROVAL, printed above.
2300 % \frontpage@bottomtext stays defined for anyone calling it directly.
2301 \vspace*{\fill}}

2302 \newcommand\coppe@hypersetup{%
2303 \begingroup
2304 % changes to \toks@ and \count@ are kept local;
2305 % it's not necessary for them, but it is usually the case
2306 % for \count1, because the first ten counters are written
2307 % to the DVI file, thus you got lucky because of PDF output
2308 \toks@={}% in this special case not necessary
2309 \count@=0 %
2310 \@whilenum\count@<\value{keywords}\do{%
2311     % * a keyword separator is not necessary,
2312     %     if there is just one keyword
2313     % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
2314     %     at least once, to get rid of the loop depended \count@
2315     \ifcase\count@ % same as \ifnum0=\count@
2316         \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
2317     \else
2318         \toks@=\expandafter\expandafter\expandafter{%
2319             \expandafter\the\expandafter\toks@
2320             \expandafter;\expandafter\space
2321             \csname CoppeKeyword:\the\count@\endcsname
2322         }%

```

```

2323 \fi
2324 \advance\count@ by 1 %
2325 }%
2326 \edef\x{\endgroup
2327 \noexpand\hypersetup{
2328 pdfkeywords={\the\toks@}%
2329 }%
2330 }%
2331 \x
2332 \hypersetup{
2333 pdfauthor={\@authname\ \@authsurn},
2334 pdftitle={\local@title},
2335 pdfsubject={\local@doctype\ de \@degree\ em \local@deptname\ da COPPE/UFRJ},
2336 pdfcreator={LaTeX with CoppeTeX toolkit},
2337 breaklinks={true},
2338 raiselinks={true},
2339 pageanchor={true},
2340 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il;” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain `TEX` command `\isundefined` (cf. the `TEX` FAQ).

```

2341 \newcommand\makecatalog{%
2342 \vspace*{\fill}
2343 \begin{center}
2344 \setlength{\fboxsep}{5mm}
2345 \framebox[120mm][c]{\makebox[5mm][c]{}%
2346 \begin{minipage}[c]{105mm}
2347 \setlength{\parindent}{5mm}
2348 \noindent\sloppy\nohyphens\@authsurn,
2349 \nohyphens\@authname\par
2350 \nohyphens{%
2351 \coppe@selecttitle/\@authname\ \@authsurn. -- \local@cityname:
2352 UFRJ/COPPE, \coppe@docyear.}\par
2353 \pageref{front:pageno},
2354 \pageref{LastPage}
2355 p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm.\par
2356 % There is an issue here. When the last entry must be split between lines,
2357 % the spacing between it and the next paragraph becomes smaller.
2358 % Should we manually introduce a fixed space? But how could we know that
2359 % a name was split? Is this happening yet?
2360 \nohyphens{%
2361 \begin{tabularx}{100mm}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}
2362 \local@advisorstring: &
2363 \count1=0
2364 \toks@={}
2365 \@whilenum \count1<\@advisor \do{%
2366 \ifcase\count1 % same as \ifnum0=\count1
2367 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2368 \expandafter\endcsname\expandafter\space%
2369 \csname CoppeAdvisorSurname:\the\count1\endcsname\}}
2370 \else
2371 \toks@=\expandafter\expandafter\expandafter{

```

```

2372         \expandafter\the\expandafter\toks@
2373         \expandafter&\expandafter\space
2374         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2375         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2376     }%
2377     \fi
2378     \advance\count1 by 1}
2379     \the\toks@
2380 \end{tabularx}}\par
2381 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
2382 UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
2383 Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
2384 \count1=0
2385 \count2=1
2386 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
2387     \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
2388     \advance\count1 by 1
2389     \advance\count2 by 1}
2390 I. \csname CoppeAdvisorSurname:0\endcsname,%
2391 \ \csname CoppeAdvisorName:0\endcsname%
2392 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
2393 II. \local@universityname, COPPE, Programa de \local@deptname.
2394 III. T{\' i}tulo.}
2395 \end{minipage}}
2396 \end{center}
2397 \vspace*{\fill}}

```

\fichacatalografica The folha adicional required from August 2026 (SiBI manual, 9th rev. ed.,
\makefolhaaditional 3.1.2.1.2 and Annex H): the CAPES collection fields plus the ficha cata-
lografica, sitting immediately after the folha de rosto and counted by no-
body. \fichacatalografica takes the PDF produced by the SiBI generator at
<<http://fichacatalografica.sibi.ufrj.br/>>, optionally with a page number.

The sheet is institutional, so it is always in Portuguese whatever the main lan-
guage of the work – and deliberately without \selectlanguage, since the brazilian
babel language need not be loaded in, say, a Spanish-main document.

```

2398 \newcommand\coppe@fichafila{}
2399 \newcommand\coppe@fichapage{1}
2400 \newcommand\fichacatalografica[2][1]{%
2401     \gdef\coppe@fichapage{#1}\gdef\coppe@fichafila{#2}}
2402 \newcommand\coppe@marcatipo[1]{%
2403     \ifthenelse{\equal{\coppe@tipoproducao}{#1}}{$\times$}{\phantom{$\times$}}}
2404 \newcommand\coppe@marcavinc[1]{%
2405     \ifthenelse{\equal{\coppe@projetovinc}{#1}}{$\times$}{\phantom{$\times$}}}
2406 \newcommand\coppe@campo[1]{%
2407     \ifthenelse{\equal{#1}{}}{\hrulefill}{#1\enspace\hrulefill}}
2408 \newcommand\coppe@checativo{%
2409     \ifthenelse{\equal{\coppe@tipoproducao}{}}{}}{%
2410     \ifthenelse{\equal{\coppe@tipoproducao}{bibliografica}}\OR
2411         \equal{\coppe@tipoproducao}{artistica}\OR
2412         \equal{\coppe@tipoproducao}{tecnologica}\OR
2413         \equal{\coppe@tipoproducao}{tecnica}}{}}{%
2414     \ClassWarning{coppe}{Unknown \string\tipoproducao{ } value
2415         '\coppe@tipoproducao'. Annex H of the SiBI manual allows only

```

```

2416         bibliografica, artistica, tecnologica or tecnica; no box will be
2417         ticked}}}}
2418 \newcommand\makefolhaadicional{%
2419     \clearpage
2420     \thispagestyle{empty}%
2421     \coppe@checativo
2422     {\singlespacing
2423     \begin{center}
2424         {\large\bfseries Informa\c c\~oes Coleta CAPES}
2425     \end{center}
2426     \vspace{7mm}
2427     \noindent Tipo de produ\c c\~ao intelectual:\par
2428     \vspace{1mm}
2429     \hspace*{10mm}(\coppe@marcatipo{bibliografica})\enspace Bibliogr\'afica\par
2430     \hspace*{10mm}(\coppe@marcatipo{artistica})\enspace Art\'istica\par
2431     \hspace*{10mm}(\coppe@marcatipo{tecnologica})\enspace Tecnol\'ogica\par
2432     \hspace*{10mm}(\coppe@marcatipo{tecnica})\enspace T\'ecnica\par
2433     \vspace{5mm}
2434     \noindent Projeto de pesquisa vinculado \'a produ\c c\~ao intelectual?\par
2435     \vspace{1mm}
2436     \hspace*{10mm}(\coppe@marcavinc{sim})\enspace Sim\par
2437     \hspace*{10mm}(\coppe@marcavinc{nao})\enspace N\~ao\par
2438     \vspace{5mm}
2439     \noindent Nome do projeto de pesquisa:\par
2440     \vspace{1mm}
2441     \noindent\coppe@campo{\coppe@nomeprojeto}\par
2442     \vspace{5mm}
2443     \noindent \'Area de concentra\c c\~ao da produ\c c\~ao intelectual:\par
2444     \vspace{1mm}
2445     \noindent\coppe@campo{\coppe@areaconc}\par
2446     \vspace{5mm}
2447     \noindent Ag\~encia(s) de fomento financiadora(s) da produ\c c\~ao
2448     intelectual:\par
2449     \vspace{1mm}
2450     \ifthenelse{\value{coppeagencias}=0}{%
2451         \noindent\hrulefill\par}{%
2452         \count1=0
2453         \@whilenum\count1<\value{coppeagencias}\do{%
2454             \noindent\hspace*{10mm}\csname CoppeAgencia:\the\count1\endcsname
2455             \ (\csname CoppeAgenciaSigla:\the\count1\endcsname)\par
2456             \advance\count1 by 1}}%
2457     \vspace{9mm}
2458     \begin{center}
2459         {\large\bfseries Ficha catalogr\'afica}\par
2460         \vspace{4mm}
2461         \ifthenelse{\equal{\coppe@fichafile}{}}{%
2462             \if@copperascunho
2463                 \makecatalog
2464             \else
2465                 \framebox[125mm][c]{\parbox[c][62mm][c]{115mm}{\centering\footnotesize
2466                     Espa\c co reservado \'a ficha catalogr\'afica.\\[1ex]
2467                     Gere-a em \texttt{fichacatalografica.sibi.ufrj.br} e inclua o
2468                     arquivo com \texttt{\string\fichacatalografica\{arquivo.pdf\}},
2469                     ou solicite-a \'a biblioteca do seu Programa.}}}%

```

```

2470     \fi}{%
2471     \includegraphics [page=\coppe@fichapage,width=125mm]{\coppe@fichafile}}%
2472 \end{center}
2473 \par}}

```

\dedication

```

2474 \newcommand\dedication[1]{
2475   \gdef\@dedic{#1}
2476   \cleardoublepage
2477   \vspace*{\fill}
2478   \begin{flushright}
2479     \begin{minipage}{60mm}
2480       \raggedleft \it \normalsize \@dedic
2481     \end{minipage}
2482   \end{flushright}}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

2483 \newenvironment{abstract}{%
2484   \cleardoublepage
2485   \thispagestyle{plain}
2486   \abstract@toptext\par
2487   \vspace*{8.6mm}
2488   \begin{center}
2489     \sloppy\nohyphens{\MakeUppercase\local@title}\par
2490     \vspace*{13.2mm}
2491     \@authname\ \@authsur \par
2492     \vspace*{7mm}
2493     \local@monthname/\coppe@docyear
2494   \end{center}\par
2495   \vspace*{1.5cm}
2496   \noindent%
2497   \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2498     \local@advisorstring: &
2499     \count1=0
2500     \toks@={}
2501     \@whilenum \count1<\@advisor \do{%
2502       \ifcase\count1 % same as \ifnum0=\count1
2503         \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2504           \expandafter\endcsname\expandafter\space%
2505           \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2506       \else
2507         \toks@=\expandafter\expandafter\expandafter{%
2508           \expandafter\the\expandafter\toks@
2509           \expandafter&\expandafter\space
2510           \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2511           \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2512       }%
2513     \fi
2514     \advance\count1 by 1}
2515   \the\toks@
2516 \end{tabularx}\par
2517 \vspace*{2mm}
2518 \noindent\local@deptstring: \local@deptname\par
2519 \vspace*{7mm}}{%

```

```

2520 \coppe@printkeywords{\coppemainstring{keywordsname}}%
2521 {CoppeKeyword:}{\value{keywords}}%
2522 \vspace*{\fill}}

```

foreignabstract (env.)

```

2523 \newenvironment{foreignabstract}{%
2524 \cleardoublepage
2525 \thispagestyle{plain}
2526 \begin{otherlanguage}{\coppe@foreignlang}
2527 \foreignabstract@toptext\par
2528 \vspace*{8.6mm}
2529 \begin{center}
2530 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
2531 \vspace*{13.2mm}
2532 \@authname\ \@authsur \par
2533 \vspace*{7mm}
2534 \foreign@monthname/\coppe@docyear
2535 \end{center}\par
2536 \vspace*{1.5cm}
2537 \noindent%
2538 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2539 \foreign@advisorstring: &
2540 \count1=0
2541 \toks@={}
2542 \@whilenum \count1<\@advisor \do{%
2543 \ifcase\count1 % same as \ifnum0=\count1
2544 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2545 \expandafter\endcsname\expandafter\space%
2546 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2547 \else
2548 \toks@=\expandafter\expandafter\expandafter{%
2549 \expandafter\the\expandafter\toks@
2550 \expandafter&\expandafter\space
2551 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2552 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2553 }%
2554 \fi
2555 \advance\count1 by 1}
2556 \the\toks@
2557 \end{tabularx}\par
2558 \vspace*{2mm}
2559 \noindent\foreign@deptstring: \foreign@deptname\par
2560 \vspace*{7mm}}{%
2561 \ifnum\value{foreignkeywords}>0
2562 \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
2563 {CoppeForeignKeyword:}{\value{foreignkeywords}}%
2564 \else
2565 \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
2566 {CoppeKeyword:}{\value{keywords}}%
2567 \fi
2568 \end{otherlanguage}
2569 \vspace*{\fill}
2570 \global\let\@author\@empty
2571 \global\let\@date\@empty

```



```

2572 \global\let\foreign@title\@empty
2573 \global\let\foreign@title\relax
2574 \global\let\local@title\@empty
2575 \global\let\local@title\relax
2576 \global\let\author\relax
2577 \global\let\author\relax
2578 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

2579 \newenvironment{brazilianabstract}{%
2580 \cleardoublepage
2581 \thispagestyle{plain}
2582 \begin{otherlanguage}{brazilian}
2583 \abstract@toptext\par
2584 \vspace*{8.6mm}
2585 \begin{center}
2586 \sloppy\nohyphens{\MakeUppercase\local@title}\par
2587 \vspace*{13.2mm}
2588 \@authname\ \@authsur \par
2589 \vspace*{7mm}
2590 \local@monthname/\coppe@docyear
2591 \end{center}\par
2592 \vspace*{1.5cm}
2593 \noindent%
2594 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2595 \local@advisorstring: &
2596 \count1=0
2597 \toks@={}
2598 \@whilenum \count1<\@advisor \do{%
2599 \ifcase\count1
2600 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2601 \expandafter\endcsname\expandafter\space%
2602 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2603 \else
2604 \toks@=\expandafter\expandafter\expandafter{%
2605 \expandafter\the\expandafter\toks@
2606 \expandafter&\expandafter\space
2607 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2608 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2609 }%
2610 \fi
2611 \advance\count1 by 1}
2612 \the\toks@
2613 \end{tabularx}\par
2614 \vspace*{2mm}
2615 \noindent\local@deptstring: \local@deptname\par
2616 \vspace*{7mm}}{%

```

```

2617 % \coppestring takes ONE argument and resolves against the current language;
2618 % this runs before \end{otherlanguage}, so \language is already brazilian.
2619 \ifnum\value{braziliankeywords}>0
2620   \coppe@printkeywords{\coppestring{keywordsname}}%
2621   {CoppeBrazilianKeyword:}{\value{braziliankeywords}}%
2622 \else
2623   \coppe@printkeywords{\coppestring{keywordsname}}%
2624   {CoppeKeyword:}{\value{keywords}}%
2625 \fi
2626 \end{otherlanguage}
2627 \vspace*{\fill}}

\listoffigures
2628 \renewcommand\listoffigures{%
2629   \coppe@hasLof
2630   \if@twocolumn
2631     \@restonecoltrue\onecolumn
2632   \else
2633     \@restonecolfalse
2634   \fi
2635   \chapter*{\listfigurename}%
2636   \addcontentsline{toc}{chapter}{\listfigurename}%
2637   \@mkboth{\MakeUppercase\listfigurename}%
2638           {\MakeUppercase\listfigurename}%
2639   \@starttoc{lof}%
2640   \if@restonecol\twocolumn\fi
2641   }

\listoftables
2642 \renewcommand\listoftables{%
2643   \if@twocolumn
2644     \@restonecoltrue\onecolumn
2645   \else
2646     \@restonecolfalse
2647   \fi
2648   \chapter*{\listtablename}%
2649   \addcontentsline{toc}{chapter}{\listtablename}%
2650   \@mkboth{%
2651     \MakeUppercase\listtablename}%
2652     {\MakeUppercase\listtablename}%
2653   \@starttoc{lot}%
2654   \if@restonecol\twocolumn\fi
2655   }

\printlosymbols
2656 \newcommand\printlosymbols{%
2657 \renewcommand\glossaryname{\listsymbolname}%
2658 \@input@{\jobname.los}}

\makelosymbols
2659 \def\makelosymbols{%
2660   \newwrite\@losfile
2661   \immediate\openout\@losfile=\jobname.syx
2662   \newcommand\symb1[3][\@bsphack\begin{group}

```

```

2663 \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
2664 \@sanitize%
2665 \@wrtos{\@tempsymb1}{##2}{##3}\typeout%
2666 {Writing index of symbols file \jobname.syx}%
2667 \let\makelosymbols\@empty%
2668 }%
2669 \@onlypreamble\makelosymbols

2670 \AtBeginDocument{%
2671 \@ifpackageloaded{hyperref}{%
2672 \newcommand\@wrtos[3]{%
2673 \protected@write\@losfile{%
2674 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2675 \endgroup%
2676 \@esphack}}{%
2677 \newcommand\@wrtos[3]{%
2678 \protected@write\@losfile{%
2679 {\string\indexentry{#1[#2] #3}{\thepage}}%
2680 \endgroup%
2681 \@esphack}}}%

```

\printloabbreviations

```

2682 \newcommand\printloabbreviations{%
2683 \renewcommand\glossaryname{\listabbreviationname}%
2684 \@input{\jobname.lab}}

```

\makeloabbreviations

```

2685 \def\makeloabbreviations{%
2686 \newwrite\@labfile
2687 \immediate\openout\@labfile=\jobname.abx
2688 \newcommand\abbrev[3][\@bsphack\beginngroup
2689 \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
2690 \@sanitize
2691 \@wrtlab{\@tempsymb1}{##2}{##3}\typeout
2692 {Writing index of abbreviations file \jobname.abx}%
2693 \let\makeloabbreviations\@empty
2694 }
2695 \@onlypreamble\makeloabbreviations

2696 \AtBeginDocument{%
2697 \@ifpackageloaded{hyperref}{%
2698 \newcommand\@wrtlab[3]{%
2699 \protected@write\@labfile{%
2700 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2701 \endgroup%
2702 \@esphack}}{%
2703 \newcommand\@wrtlab[3]{%
2704 \protected@write\@labfile{%
2705 {\string\indexentry{#1[#2] #3}{\thepage}}%
2706 \endgroup%
2707 \@esphack}}}%

```

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
 \sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in

the list of abbreviations and acronyms (through `\abbrev`, so `\makeloabbreviations` must be active), then just “SIGLA” on later uses. `\sigla*{key}` forces the full form. The first-use flag is set globally so it survives groups (floats, footnotes).

```

2708 \newcommand\newsigla[3]{%
2709   \@namedef{coppe@sigla@s@#1}{#2}%
2710   \@namedef{coppe@sigla@l@#1}{#3}}
2711 \newcommand\coppe@sigla@reg[1]{%
2712   \ifundefined{abbrev}{}%
2713     {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}%
2714 \newcommand\coppe@sigla@full[1]{%
2715   \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
2716 \newcommand\coppe@sigla@undef[1]{%
2717   \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
2718   \textbf{???#1??}}
2719 \newcommand\coppe@sigla@first[1]{%
2720   \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
2721 \newcommand\coppe@sigla@norm[1]{%
2722   \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
2723   {\ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
2724     {\@nameuse{coppe@sigla@s@#1}}}%
2725 \newcommand\coppe@sigla@star[1]{%
2726   \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
2727   {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}%
2728 \newcommand\sigla{\ifstar\coppe@sigla@star\coppe@sigla@norm}

2729 %%% \AtBeginDocument{%
2730 %%% \ifpackageloaded{hyperref}{%
2731 %%%   \def\@wrlab#1#2{%
2732 %%%     \protected@write\@labfile{%
2733 %%%       {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
2734 %%%     \endgroup
2735 %%%     \@esphack}}{%
2736 %%%   \def\@wrlab#1#2{%
2737 %%%     \protected@write\@labfile{%
2738 %%%       {\string\indexentry{[#1] #2}{\arabic{page}}}%
2739 %%%     \endgroup
2740 %%%     \@esphack}}}

2741 % Catalog-label writers. Two language-related traps must be avoided here:
2742 % 1. The label NAMES contain ':' and '.', which babel makes active in
2743 %    some languages (e.g. Brazilian's "shorthand" set). If the write
2744 %    argument is tokenised while one of those is active, pdfTeX aborts
2745 %    with "<inserted text> \endwrite" because the shorthand expansion
2746 %    injects unbalanced braces. \detokenize emits its argument as
2747 %    catcode-12 characters regardless of the current activation, so the
2748 %    label NAMES survive whatever language is current at writer time.
2749 % 2. Spanish babel redefines \@roman (and therefore \roman{...}) to
2750 %    produce small-caps roman numerals via \es@scroman, whose expansion
2751 %    contains an \uppercase group that is unbalanced inside \write. The
2752 %    same crash signature appears. The fix is to bypass LaTeX's
2753 %    \roman/\arabic counter wrappers entirely and use the raw e-TeX
2754 %    primitives \romannumeral and \number directly on the \c@page count
2755 %    register --- both are fully expandable, language-agnostic, and
2756 %    always produce plain ASCII digits or lowercase roman letters.
2757 %    \Roman{page} is left in place because babel-spanish only redefines

```

```

2758 %      \@Roman under the (non-default) ucroman option.
2759 \AtBeginDocument{%
2760   \ltx@ifpackageloaded{hyperref}{%
2761     \def\coppe@bibEnd{%
2762       \immediate\write\@auxout{%
2763         \string\newlabel{\detokenize{bib:end}}%
2764         {{{\number\c@page}}{\detokenize{page.}\number\c@page}}}}}%
2765   \def\coppe@bibBegin{%
2766     \immediate\write\@auxout{%
2767       \string\newlabel{\detokenize{bib:begin}}%
2768       {{{\number\c@page}}{\detokenize{page.}\number\c@page}}}}}%
2769   \def\coppe@mainBegin{%
2770     \immediate\write\@auxout{%
2771       \string\newlabel{\detokenize{front:pageno}}%
2772       {{{\Roman{page}}{\detokenize{page.}\romannumeral\c@page}}}}}%
2773   \def\coppe@hasLof{%
2774     \immediate\write\@auxout{%
2775       \string\newlabel{\detokenize{cat:lofflag}}%
2776       {{{\detokenize{:~il.}}{\detokenize{page.}\romannumeral\c@page}}}}}%
2777 }{%
2778   \def\coppe@bibEnd{%
2779     \immediate\write\@auxout{%
2780       \string\newlabel{\detokenize{bib:end}}{{{ {\number\c@page}}}}}%
2781   \def\coppe@bibBegin{%
2782     \immediate\write\@auxout{%
2783       \string\newlabel{\detokenize{bib:begin}}{{{ {\number\c@page}}}}}%
2784   \def\coppe@mainBegin{%
2785     \immediate\write\@auxout{%
2786       \string\newlabel{\detokenize{front:pageno}}{{{ {\Roman{page}}}}}}}%
2787   \def\coppe@hasLof{%
2788     \immediate\write\@auxout{%
2789       \string\newlabel{\detokenize{cat:lofflag}}{{{ {\detokenize{:~il.}}}}}}}%
2790 }%
2791 }
2792 % The reference list is now produced by biblatex's \printbibliography (see the
2793 % bibliography setup after babel, above). The former thebibliography
2794 % redefinition and its \bibindent are no longer used and have been removed; the
2795 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
2796 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (*env.*)

```

2797 \newlength{\recuolongquote}
2798 \setlength{\recuolongquote}{4cm}
2799
2800 \newenvironment*{longquote}[1][default]{%
2801   \begin{list}{}{%
2802     \setlength{\leftmargin}{\recuolongquote}%
2803     \setlength{\rightmargin}{0pt}%
2804     \setlength{\itemindent}{0pt}%
2805     \setlength{\listparindent}{0pt}%
2806     \setlength{\parsep}{0pt}%
2807     \setlength{\topsep}{\baselineskip}%
2808   }%
2809   \item\relax

```

```

2810 \footnotesize
2811 \singlespacing
2812 \ifthenelse{\not\equal{#1}{default}}{%
2813 {\itshape\selectlanguage{#1}}%
2814 }%
2815 \ignorespaces
2816 }{%
2817 \end{list}%
2818 \ignorespacesafterend
2819 }

2820 \newenvironment{theglossary}{%
2821 \if@twocolumn%
2822 \@restonecoltrue\onecolumn%
2823 \else%
2824 \@restonecolfalse%
2825 \fi%
2826 \@mkboth{\MakeUppercase\glossaryname}%
2827 {\MakeUppercase\glossaryname}%
2828 \chapter*{\glossaryname}%
2829 \addcontentsline{toc}{chapter}{\glossaryname}
2830 \list{}
2831 {\setlength{\listparindent}{0in}%
2832 \setlength{\labelwidth}{1.0in}%
2833 \setlength{\leftmargin}{1.5in}%
2834 \setlength{\labelsep}{0.5in}%
2835 \setlength{\itemindent}{0in}}%
2836 \sloppy}%
2837 {\if@restonecol\twocolumn\fi%
2838 \endlist}
2839 %
2840 \renewenvironment{theindex}{%
2841 \if@twocolumn
2842 \@restonecolfalse
2843 \else
2844 \@restonecoltrue
2845 \fi
2846 \twocolumn[\@makeschapterhead{\indexname}]%
2847 \@mkboth{\MakeUppercase\indexname}%
2848 {\MakeUppercase\indexname}%
2849 \thispagestyle{plain}\parindent\z@
2850 \addcontentsline{toc}{chapter}{\indexname}
2851 \parskip\z@ \@plus .3\p@\relax
2852 \columnseprule \z@
2853 \columnsep 35\p@
2854 \let\item\@idxitem}
2855 {\if@restonecol\onecolumn\else\clearpage\fi}
2856 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
2857 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
2858 \newcommand\glossaryname{\coppemainstring{glossary}}
2859 %
2860 \newcommand\local@advisorstring{Orientador}
2861 \newcommand\foreign@advisorstring{Advisor}
2862 % Fed by the string dispatcher instead of a hard-coded Portuguese literal.

```

```

2863 % Expansion is lazy (it happens inside \makefrontpage), so \coppe@mainlang is
2864 % already set by the time this is used.
2865 \ifthenelse{\boolean{maledoc}}{%
2866   \newcommand\local@approvedname{\coppemainstring{approvedmale}}%
2867 }{%
2868   \newcommand\local@approvedname{\coppemainstring{approvedfemale}}%
2869 }
2870 \ifthenelse{\boolean{maledoc}}{%
2871   \newcommand\local@approvedonname{\coppemainstring{approvedonmale}}%
2872 }{%
2873   \newcommand\local@approvedonname{\coppemainstring{approvedonfemale}}%
2874 }
2875 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
2876 \newcommand\local@deptstring{Programa}
2877 \newcommand\foreign@deptstring{Department}
2878 \newcommand\local@cityname{Rio de Janeiro}
2879 \newcommand\local@statename{RJ}
2880 \newcommand\local@countryname{Brasil}
2881 %
2882 % 3.1.2.1.1(e) of the 2026 manual requires the natureza, the objetivo, the
2883 % name of the institution AND the area de concentracao in this block. The area
2884 % was missing; it is appended as its own sentence when \areaconcentracao was
2885 % given, and simply omitted otherwise.
2886 \newcommand\frontcover@maintext{
2887   \sloppy\nohyphens{\local@doctype\ de \@degreename\
2888   \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2889   ao Programa de P{\ ' o}s-gradua{\c c}{\~ a}o em \local@deptname,
2890   COPPE, da \local@universityname, como parte dos requisitos
2891   necess{\ ' a}rios {\ ' a} obten{\c c}{\~ a}o do t{\ ' i}tulo de
2892   \local@degname\ em \local@deptname.}%
2893   \ifthenelse{\equal{\coppe@areaconc}{}}{}{%
2894     \par\vspace{1ex}%
2895     \sloppy\nohyphens{\ ' Area de concentra{\c c}{\~ a}o: \coppe@areaconc.}}%
2896 }
2897 %
2898 \newcommand\frontpage@maintext{
2899   \noindent {\MakeUppercase\local@doctype}
2900   \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
2901   \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
2902   DE P{\ ' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
2903   UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
2904   NECESS{\ ' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
2905   {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
2906   {\MakeUppercase\local@deptname.\par}}%
2907 }
2908 %
2909 \newcommand\frontpage@bottomtext{%
2910   \begin{center}
2911     {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
2912     {\MakeUppercase\local@monthname\ DE \coppe@docyear}
2913   \end{center}%
2914 }
2915 %
2916 \newcommand\abstract@toptext{%

```

```

2917 \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
2918 \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2919 \sloppy\nohyphens{{\` a} COPPE/UFRJ como parte dos requisitos
2920 necess{\` a}rios para a obten{\c c}{\~ a}o do grau de
2921 \local@degname\ em Ci{\` e}ncias (\@degree)}
2922 }
2923 \newcommand\foreignabstract@toptext{%
2924 \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
2925 COPPE/UFRJ as a partial fulfillment of the requirements for the
2926 degree of \foreign@degname\ of Science (\@degree)}
2927 }
2928 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before **hyperref** patches all the cross-reference and TOC plumbing.
- **hyperref** – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, `list-of-*` commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the **coppe** class redefines (e.g. `\@chapter`, `\listofquadros`).
- **biblatex** – must come after **hyperref** so its citation links and back-references hook into **hyperref** correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```

2929 \RequirePackage{setspace}
2930 % Apply the default line spacing now that setspace is available. The flag was
2931 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
2932 \if@coppedoublespace\doublespacing\else\onehalfspacing\fi
2933 % pdfx MUST precede hyperref -- it loads hyperref itself, with the options
2934 % PDF/A needs, and cannot fix up an already-loaded one. Level a-2b rather than
2935 % a-1b (the "ISO 19005-1" of Annex I): a-1b forbids transparency, which
2936 % tcolorbox and several graphics packages produce, so a-1b would fail on
2937 % documents that are otherwise perfectly fine. a-2b is ISO 19005-2, accepted by
2938 % Pantheon and by the SiBI's own instructions, which also list PDF/A-2b.
2939 \if@coppepdfa
2940 \RequirePackage[a-2b]{pdfx}
2941 % pdfTeX stamps /PTEX.Fullbanner into the document information dictionary.
2942 % It is not one of the keys ISO 19005-2 6.6.2.3.2 maps to an XMP property, and
2943 % no extension schema declares it, so it is a DocInfo entry with no XMP
2944 % counterpart -- a finding a PDF/A checker has every reason to raise, and one
2945 % that costs nothing to remove. -1 sets every suppression bit.
2946 \pdfsuppressptexinfo=-1
2947 % pdfx takes the XMP metadata from \jobname.xmpdata, which it reads when it is
2948 % LOADED -- before the preamble has run, so before \title, \author and
2949 % \keyword exist. The file is therefore WRITTEN by the class and picked up on
2950 % the NEXT run. The build already runs pdflatex three times for cross-

```



```

2951 % references, so the deposited PDF always carries current metadata; only a
2952 % first-ever compilation of a brand-new file emits pdfx's "no \jobname.xmpdata"
2953 % warning and lacks dc:title.
2954 %
2955 % The write happens at the END of the document, not at \begin{document}. This
2956 % class has always documented -- and every example follows it -- that \title,
2957 % \author, \department and \keyword go AFTER \begin{document}, next to
2958 % \maketitle. A write at \begin{document} therefore ran before any of them
2959 % existed and died with "Undefined control sequence" on \@authname, so the
2960 % pdfa option was unusable in exactly the style the manual teaches. It only
2961 % ever worked for a document that set its metadata in the preamble. At the end
2962 % everything is set whichever style the author chose, and \immediate still puts
2963 % the file on disk in time for the next run.
2964 %
2965 % The hook is enddocument/afteraux, and the stream is \@mainaux rather than one
2966 % of our own. TeX has sixteen output streams and example.tex was already using
2967 % all of them (toc, lof, lot, loa, lol, loq, los, idx, glo, aux, bcf, ...), so
2968 % \newwrite here failed with "No room for a new \write" on precisely the
2969 % document that most needs validating. enddocument/afteraux fires after LaTeX
2970 % has closed the .aux, which frees that stream for the one \immediate\openout
2971 % we need and costs the document nothing. Kernels older than 2020-10-01 have no
2972 % such hook; there the class falls back to \AtEndDocument with a stream of its
2973 % own, which is the pre-existing behaviour.
2974 %
2975 % Every field is guarded, and the guards are not cosmetic. \coppe@selecttitle
2976 % is \@nameuse{coppe@title@<lang>}, and \csname on a name that does not exist
2977 % yet MAKES it \relax: unexpandable, so \write puts the control sequence NAME
2978 % into the file instead of the title. The .xmpdata then carried the literal
2979 % \coppe@title@brazilian, pdfx read it back on the following run, and the
2980 % document stopped compiling at all -- a single bad run bricked the file until
2981 % someone deleted the .xmpdata by hand. A field that is not set is now simply
2982 % omitted, which pdfx handles, instead of poisoning the next run.
2983 %
2984 % The writes must be \immediate. \protected@write defers to the shipout, and
2985 % the stream is closed here and now, so deferred writes would land nowhere --
2986 % which is exactly what happened first time round: a 0-byte .xmpdata.
2987 % \let\protect\string inside the group gives the protection that
2988 % \protected@write would have given: accents and other robust commands are
2989 % written literally instead of being expanded. pdfx reads the .xmpdata as
2990 % LaTeX, so \'a arrives there intact.
2991 \ifdefined\AddToHook
2992   \let\coppe@xmpout\@mainaux
2993   \def\coppe@xmphook{\AddToHook{enddocument/afteraux}}%
2994 \else
2995   \newwrite\coppe@xmpout
2996   \let\coppe@xmphook\AtEndDocument
2997 \fi
2998 \coppe@xmphook{%
2999   \gdef\coppe@xmpkeys{%
3000     \count@=0
3001     \@whilenum\count@<\value{keywords}\do{%
3002       \ifnum\count@>0
3003         \g@addto@macro\coppe@xmpkeys{\string\sep\space}%
3004       \fi

```

```

3005     \protected@edef\coppe@tmp{\csname CoppeKeyword:\the\count@\endcsname}%
3006     \expandafter\g@addto@macro\expandafter\coppe@xmpkeys
3007     \expandafter{\coppe@tmp}%
3008     \advance\count@ by 1}%
3009 \immediate\openout\coppe@xmpout=\jobname.xmpdata
3010 \begingroup
3011     \let\protect\string
3012     \@ifundefined{coppe@title@\coppe@mainlang}{\}%
3013     \immediate\write\coppe@xmpout{\string\Title{\coppe@selecttitle}}}%
3014     \@ifundefined{coppe@authorname}{\}%
3015     \immediate\write\coppe@xmpout{%
3016         \string\Author{\@authname\space\@authsurname}}%
3017     \immediate\write\coppe@xmpout{\string\Keywords{\coppe@xmpkeys}}%
3018     \ifthenelse{equal{\coppe@deptcode}{}}{\}%
3019     \immediate\write\coppe@xmpout{\string\Subject{\local@doctype\space de
3020         \@degree name. Programa de \meta@deptname\space(\coppe@deptcode),
3021         COPPE/UFRJ}}}%
3022     \immediate\write\coppe@xmpout{\string\Publisher{\local@universityname}}%
3023     \@ifundefined{coppe@docyear}{\}%
3024     \immediate\write\coppe@xmpout{\string\Date{\coppe@docyear}}}%
3025 \endgroup
3026 \immediate\closeout\coppe@xmpout}%
3027 \fi
3028 \RequirePackage{hyperref}
3029 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
3030 % former natbib+bibtex setup. The numbers class option selects the numeric
3031 % citation style; otherwise the author-date default is used. natbib=true keeps
3032 % \citet/\citep available for users migrating from natbib.
3033 \if@coppenumeric
3034     \RequirePackage[backend=biber,datamodel=coppe,%
3035         bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
3036 \else
3037     \RequirePackage[backend=biber,datamodel=coppe,%
3038         bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
3039 \fi
3040 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
3041 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
3042 % "Bibliografia" (babel's default \bibname). The title is rendered through
3043 % \coppestring so it follows the current main language in BOTH the chapter
3044 % heading and the .toc entry (biblatex's \bibstring proved unreliable across
3045 % the .toc write, and babel's \bibname caption override was equally fragile).
3046 \defbibheading{bibliography}{%
3047     \chapter*{\coppestring{references}}%
3048     \addcontentsline{toc}{chapter}{\coppestring{references}}}
3049 % The catalog page range (bib:begin/bib:end) was written by the old
3050 % thebibliography redefinition, which biblatex's \printbibliography does not
3051 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
3052 % begin label uses that hook and the end label is emitted by wrapping
3053 % \printbibliography to write it just after the reference list.
3054 \AtBeginBibliography{\coppe@bibBegin}
3055 \let\coppe@orig@printbibliography\printbibliography
3056 \renewcommand\printbibliography[1][\]%
3057     \coppe@orig@printbibliography[#1]%
3058 \coppe@bibEnd}

```

```

3059 </class>
3060 <*glossary>
3061 actual '='
3062 quote '!'
3063 level '>'
3064 %%% delim_0 " , p. "
3065 delim_0 "\\dotfill "
3066 lethead_flag 0
3067 headings_flag 0
3068 preamble
3069 "\n\\begin{theglossary}\n \\makeatletter"
3070 postamble
3071 "\n \\end{theglossary}\n"
3072 </glossary>
3073 <*class>
3074 \newcommand{\annex}{
3075     \renewcommand{\appendixname}{\coppestring{annex}}%
3076     \appendix
3077     \@coppeannextrue
3078 }
3079 </class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

3080 <*dbx>
3081 \ProvidesFile{coppe.dbx}[2026/05/28 v4.0 CoppeTeX ABNT datamodel]
3082 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
3083 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
3084 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
3085 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
3086 % performance) are already in biblatex's standard datamodel.
3087 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
3088 % Custom fields.
3089 \DeclareDatamodelFields[type=field,datatype=literal]{course} % "(... em <course>)"
3090 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
3091 \DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
3092 \DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
3093 \DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
3094 \DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video
3095 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
3096 \DeclareDatamodelFields[type=field,datatype=literal]{caderno} % newspaper section
3097 \DeclareDatamodelFields[type=field,datatype=literal]{relator} % jurisprudence rapporteur
3098 \DeclareDatamodelEntryfields{course,coppedegree}
3099 \DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
3100 {artist,platform,director,producer,version}
3101 \DeclareDatamodelEntryfields[map]{scale}
3102 \DeclareDatamodelEntryfields[article]{caderno}
3103 \DeclareDatamodelEntryfields[jurisdiction]{relator}
3104 </dbx>
3105 <*bbx>
3106 \ProvidesFile{coppe.bbx}[2026/05/28 v4.0 CoppeTeX ABNT bibliography style]
3107 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in

```

```

3108 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
3109 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
3110 % so load setspace here as well (idempotent).
3111 \RequirePackage{setspace}
3112 % ABNT reference list keeps the date at the end (imprenta), same for author-date
3113 % and numeric modes, so we derive from the standard style, not authoryear.
3114 \RequireBibliographyStyle{standard}
3115
3116 % Note: standard style never dashes repeated authors, which already matches the
3117 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
3118 \ExecuteBibliographyOptions{%
3119     maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
3120     giveninits=false,
3121     uniquename=false,
3122     uniquelist=false,
3123     labeldateparts=true, % needed for author-date extradate (1999a/1999b)
3124     sorting=nyt,
3125 }
3126
3127 % Language strings (Disponível em / Acesso em / In, etc.)
3128 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
3129 \DeclareLanguageMapping{english}{english-coppe}
3130
3131 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
3132 \DeclareNameAlias{author}{family-given}
3133 \DeclareNameAlias{editor}{family-given}
3134 \DeclareNameAlias{translator}{family-given}
3135 \DeclareNameAlias{default}{family-given}
3136 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
3137 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
3138
3139 % ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
3140 % organizer prints the kind of responsibility in parentheses after the
3141 % name, abbreviated, lowercase and in the singular even for several
3142 % organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
3143 % standard style prints ", Org." (comma, no parentheses) when the editor
3144 % takes the author slot, so override editor+others to wrap the role string
3145 % in "(...)". The host position ("In: NAME (org.)") already parenthesises
3146 % the role via coppe:bookeditor below. The role word itself is the
3147 % lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
3148 % packs (brazilian-coppe.lbx et al.).
3149 \renewbibmacro*{editor+others}{%
3150     \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
3151         {\printnames{editor}%
3152          \setunit*{\addspace}%
3153          \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
3154          \clearname{editor}}
3155     {}
3156 % Same parenthesised role for drivers that take the editor through the plain
3157 % editor macro instead of editor+others (e.g. @manual).
3158 \renewbibmacro*{editor}{%
3159     \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
3160         {\printnames{editor}%
3161          \setunit*{\addspace}%

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3162 \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3163 \clearname{editor}}
3164 {}
3165
3166 % Uppercase the surname (family + prefix) only inside the bibliography list;
3167 % citations keep normal caps (post-2020 rule).
3168 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
3169 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
3170 % field names; the month stays abbreviated via \mkbibmonth.
3171 \protected\def\coppe@abntdate#1#2#3{%
3172 \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
3173 \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
3174 \iffieldundef{#1}{\stripzeros{\thefield{#1}}}%
3175 \AtBeginBibliography{%
3176 \singlespacing % entries single-spaced internally (2.4)
3177 \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
3178 \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
3179 \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
3180 \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
3181 \let\mkbibdateshort\coppe@abntdate
3182 }
3183
3184 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
3185 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
3186 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
3187 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
3188 % article AND the next word are uppercased; otherwise just the first word. The
3189 % delimited macros split off words; \coppeUCkeep strips the trailing-space
3190 % sentinel so no spurious space is left before the following punctuation.
3191 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
3192 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
3193 \coppeUCifart{#1}
3194 {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
3195 {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
3196 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
3197 \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
3198 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
3199 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
3200 \protected\def\coppeUCifart#1{%
3201 \begingroup
3202 \lowercase{\def\coppeUC@w{#1}}%
3203 \edef\coppeUC@w{\coppeUC@w}%
3204 \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,}%
3205 \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
3206 \endgroup}
3207 \DeclareFieldFormat{title}{%
3208 \ifboolexpr{ test {\ifnameundef{labelname}} }
3209 {#1}%
3210 {\mkbibbold{#1}}}
3211 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
3212 % format receives the RAW title text (the title format above only sees the
3213 % field-printing commands), so \coppeUCtitle can split off and uppercase the
3214 % first word here.
3215 \DeclareFieldFormat{titlecase}{%

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3216 \ifboolexpr{ test {\ifnameundef{labelname}} }
3217   {\coppeUCtitle{#1}}%
3218   {#1}}
3219 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{
3220 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
3221 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
3222 % the title of a whole document, so override those back to bold.
3223 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
3224 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
3225 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
3226 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
3227 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
3228 \DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
3229 \DeclareFieldFormat{maintitle}{\mkbibbold{#1}}
3230
3231 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
3232 \DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
3233 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
3234 \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
3235
3236 % --- Periodical volume / number prefixes ---
3237 \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
3238 \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
3239
3240 % --- "In:" before host title (incollection/inproceedings/inbook) ---
3241 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
3242
3243 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
3244 \DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}
3245
3246 % --- ABNT article driver: no "In."; Journal, Local, v., n., p., date, DOI/URL.
3247 %   Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
3248 %   differ: the newspaper name (journaltitle) and city, then either
3249 %       "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
3250 %       "p. <pages>, <date>" (page before date, when none),
3251 %   per NBR 6023 §4.2.3.6. ---
3252 \DeclareBibliographyDriver{article}{%
3253   \usebibmacro{bibindex}%
3254   \usebibmacro{begentry}%
3255   \usebibmacro{author/translator+others}%
3256   \setunit{\labelnamepunct}\newblock
3257   \usebibmacro{title}%
3258   \newunit\newblock
3259   \printfield{journaltitle}%
3260   \setunit{\addcomma\space}\printlist{location}%
3261   \iffieldequalstr{entrysubtype}{newspaper}
3262     {\iffieldundef{caderno}
3263       {\setunit{\addcomma\space}\printfield{pages}%
3264        \setunit{\addcomma\space}\printdate}%
3265       {\setunit{\addcomma\space}\printdate%
3266        \setunit{\adddot\space}\printfield{caderno}%
3267        \setunit{\addcomma\space}\printfield{pages}}}%
3268   {\setunit{\addcomma\space}\printfield{volume}%
3269    \setunit{\addcomma\space}\printfield{number}%

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3270 \setunit{\addcomma\space}\printfield{pages}%
3271 \setunit{\addcomma\space}\printdate}%
3272 \newunit\newblock
3273 \printfield{doi}%
3274 \newunit\newblock
3275 \usebibmacro{url+urldate}%
3276 \usebibmacro{finentry}}
3277
3278 % --- incollection: ABNT puts the host organizer before the booktitle, as
3279 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
3280 % p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
3281 \newbibmacro*{coppe:bookeditor}{%
3282 \ifnameundef{editor}
3283 {}
3284 {\printnames{editor}%
3285 \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3286 \clearname{editor}%
3287 \setunit{\adddot\space}}}
3288 \newbibmacro*{coppe:partdriver}{%
3289 \usebibmacro{bibindex}\usebibmacro{begentry}%
3290 \usebibmacro{author/translator+others}%
3291 \setunit{\labelnamepunct}\newblock
3292 \usebibmacro{title}%
3293 \newunit\newblock
3294 \usebibmacro{in:}%
3295 \usebibmacro{coppe:bookeditor}%
3296 \printfield{booktitle}%
3297 \newunit\newblock
3298 \printfield{edition}%
3299 \newunit\newblock
3300 \printlist{location}%
3301 \setunit*{\addcolon\space}\printlist{publisher}%
3302 \setunit*{\addcomma\space}\printdate%
3303 \newunit\newblock
3304 \printfield{pages}%
3305 \setunit{\addcomma\space}\printfield{chapter}%
3306 \newunit\newblock
3307 \usebibmacro{url+urldate}%
3308 \usebibmacro{finentry}}
3309 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
3310 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
3311
3312 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
3313 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
3314
3315 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
3316 \DeclareFieldFormat{course}{\mkbibparens{\coppestring{in:}\addspace#1}}
3317 \DeclareBibliographyDriver{thesis}{%
3318 \usebibmacro{bibindex}\usebibmacro{begentry}%
3319 \usebibmacro{author}%
3320 \setunit{\labelnamepunct}\newblock
3321 \usebibmacro{title}%
3322 \newunit\newblock
3323 \printfield{type}\setunit*{\addspace}\printfield{course}%

```

```

3324 \newunit\newblock
3325 \printlist{institution}%
3326 \setunit*{\addcomma\space}\printlist{location}%
3327 \setunit*{\addcomma\space}\printdate%
3328 \newunit\newblock
3329 \printfield{pagetotal}%
3330 \newunit\newblock
3331 \printfield{note}%
3332 \newunit\newblock
3333 \usebibmacro{url+urldate}%
3334 \usebibmacro{finentry}}
3335 % --- Games / audiovisual: render the custom fields (artist, platform,
3336 % director, producer, version) the generic misc driver ignores. ---
3337 \newbibmacro*{coppe:media}{%
3338 \iflistundef{artist}{\setunit{\addperiod\space}%
3339 \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
3340 \iffieldundef{platform}{\setunit{\addperiod\space}%
3341 \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}}%
3342 \iflistundef{director}{\setunit{\addperiod\space}%
3343 \printtext{\coppestring{directedby}\addcolon\space}\printlist{director}}%
3344 \iflistundef{producer}{\setunit{\addperiod\space}%
3345 \printtext{\coppestring{producedby}\addcolon\space}\printlist{producer}}}%
3346 \newbibmacro*{coppe:mediadriver}{%
3347 \usebibmacro{bibindex}\usebibmacro{begentry}%
3348 \usebibmacro{author/translator+others}%
3349 \setunit{\labelnamepunct}\newblock
3350 \usebibmacro{title}%
3351 \newunit\newblock
3352 \usebibmacro{coppe:media}%
3353 \newunit\newblock
3354 \printlist{publisher}%
3355 \setunit*{\addcomma\space}\printlist{location}%
3356 \setunit*{\addcomma\space}\printdate%
3357 \newunit\newblock
3358 \printfield{version}%
3359 \newunit\newblock
3360 \printfield{note}%
3361 \newunit\newblock
3362 \usebibmacro{url+urldate}%
3363 \usebibmacro{finentry}}
3364 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
3365 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
3366 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
3367 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
3368 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
3369 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
3370 % imprenta) plus the mandatory scale ("Escala 1:N").
3371 \DeclareFieldFormat{scale}{\bibstring{coppescale}\adnbspace#1}
3372 \DeclareBibliographyDriver{map}{%
3373 \usebibmacro{bibindex}\usebibmacro{begentry}%
3374 \usebibmacro{author/translator+others}%
3375 \setunit{\labelnamepunct}\newblock
3376 \usebibmacro{title}%
3377 \newunit\newblock

```



```

3378 \printlist{location}%
3379 \setunit*{\addcolon\space}\printlist{publisher}%
3380 \setunit*{\addcomma\space}\printdate%
3381 \newunit\newblock
3382 \printfield{scale}%
3383 \newunit\newblock
3384 \usebibmacro{url+urldate}%
3385 \usebibmacro{finentry}}
3386 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
3387 % number. date. The holder (depositante) is printed in normal case, not the
3388 % ALL-CAPS family treatment the bibliography gives the leading author.
3389 \newbibmacro*{coppe:depositor}{%
3390 \ifnameundef{holder}
3391 {}
3392 {\printtext{\bibstring{depositor}\addcolon\space}%
3393 \begingroup
3394 \def\mkbibnamefamily##1{##1}%
3395 \def\mkbibnameprefix##1{##1}%
3396 \printnames{holder}%
3397 \endgroup}}
3398 \DeclareBibliographyDriver{patent}{%
3399 \usebibmacro{bibindex}%
3400 \usebibmacro{begentry}%
3401 \usebibmacro{author/translator+others}%
3402 \setunit{\labelnamepunct}\newblock
3403 \usebibmacro{title}%
3404 \newunit\newblock
3405 \usebibmacro{coppe:depositor}%
3406 \newunit\newblock
3407 \printfield{type}%
3408 \setunit*{\addspace}\printfield{number}%
3409 \newunit\newblock
3410 \printdate%
3411 \newunit\newblock
3412 \usebibmacro{url+urldate}%
3413 \usebibmacro{finentry}}
3414 % Exotic ABNT types that biblatex's standard style leaves without a driver:
3415 % give each a baseline (book-like for published docs: norma/partitura; misc for
3416 % the rest) so they print. Per-type ABNT refinements can follow later.
3417 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
3418 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
3419 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
3420 % the law id + ementa (plain, not bold), then location and date.
3421 \newbibmacro*{coppe:legaldriver}{%
3422 \usebibmacro{bibindex}\usebibmacro{begentry}%
3423 \usebibmacro{author/translator+others}%
3424 \setunit{\labelnamepunct}\newblock
3425 \usebibmacro{title}%
3426 \newunit\newblock
3427 % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
3428 \iffieldundef{relator}{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\newblock}
3429 \printlist{location}%
3430 \setunit{\addcomma\space}\printdate%
3431 \newunit\newblock

```

```

3432 \usebibmacro{url+urldate}%
3433 \usebibmacro{finentry}}
3434 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
3435 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
3436 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
3437 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
3438 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
3439 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
3440 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
3441 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
3442 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
3443 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
3444
3445 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
3446 \DeclareSourcemap{%
3447   \maps[datatype=bibtex]{%
3448     % entry-type PT synonyms -> canonical English
3449     \map{step[typesource=livro,typetarget=book]}
3450     \map{step[typesource=artigo,typetarget=article]}
3451     \map{step[typesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
3452     \map{step[typesource=partedelivro,typetarget=incollection]}
3453     \map{step[typesource=emlivro,typetarget=inbook]}
3454     \map{step[typesource=verbete,typetarget=inreference]}
3455     \map{step[typesource=anais,typetarget=proceedings]}
3456     \map{step[typesource=trabalhodeevento,typetarget=inproceedings]}
3457     \map{step[typesource=periodico,typetarget=periodical]}
3458     \map{step[typesource=relatorio,typetarget=report]}
3459     \map{step[typesource=relatoriotecnico,typetarget=report]}
3460     \map{step[typesource>manual,typetarget>manual]}
3461     \map{step[typesource=correspondencia,typetarget=letter]}
3462     \map{step[typesource=patente,typetarget=patent]}
3463     \map{step[typesource=diverso,typetarget=misc]}
3464     \map{step[typesource=jogo,typetarget=game]}
3465     \map{step[typesource=jogoeletronico,typetarget=videogame]}
3466     \map{step[typesource=jogodetabuleiro,typetarget=boardgame]}
3467     \map{step[typesource=filme,typetarget=movie]}
3468     \map{step[typesource=serietv,typetarget=tvshow]}
3469     \map{step[typesource=programadetelevisao,typetarget=tvshow]}
3470     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
3471     % except map (declared in coppe.dbx).
3472     \map{step[typesource=legislacao,typetarget=legislation]}
3473     \map{step[typesource=atonomativo,typetarget=legislation]}
3474     \map{step[typesource=jurisprudencia,typetarget=jurisdiction]}
3475     \map{step[typesource=documentojuridico,typetarget=legal]}
3476     \map{step[typesource=documentocivil,typetarget=legal]}
3477     \map{step[typesource=norma,typetarget=standard]}
3478     \map{step[typesource=partitura,typetarget=music]}
3479     \map{step[typesource=iconografico,typetarget=image]}
3480     \map{step[typesource=obradearte,typetarget=artwork]}
3481     \map{step[typesource=tridimensional,typetarget=artwork]}
3482     \map{step[typesource=cartografico,typetarget=map]}
3483     \map{step[typesource=mapa,typetarget=map]}
3484     \map{step[typesource=audiovisual,typetarget=video]}
3485     \map{step[typesource=conjuntodedados,typetarget=dataset]}

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```

3486 \map{step[typesource=fasciculo,typetarget=supperiodical]}
3487 \map{step[typesource=trabalhoapresentado,typetarget=unpublished]}
3488 % thesis family: retype to thesis and preset the (babel-aware) type word
3489 \map{step[typesource=tese,typetarget=thesis]}
3490 \map{step[typesource=dissertacao,typetarget=thesis]}
3491 \map{step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
3492 \map{step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fieldv
3493 \map{step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
3494 \map{step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
3495 \map{step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
3496 \map{step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
3497 \map{step[typesource=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
3498 \map{step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
3499 % field PT synonyms (unaccented) -> canonical English
3500 \map{step[fieldsource=periodico,fieldtarget=journaltitle]}
3501 \map{step[fieldsource=revista,fieldtarget=journaltitle]}
3502 \map{step[fieldsource=jornal,fieldtarget=journaltitle]}
3503 \map{step[fieldsource=titulo,fieldtarget=title]}
3504 \map{step[fieldsource=subtitulo,fieldtarget=subtitle]}
3505 \map{step[fieldsource=autor,fieldtarget=author]}
3506 \map{step[fieldsource=editora,fieldtarget=publisher]}
3507 \map{step[fieldsource=local,fieldtarget=location]}
3508 \map{step[fieldsource=ano,fieldtarget=year]}
3509 \map{step[fieldsource=data,fieldtarget=date]}
3510 \map{step[fieldsource=edicao,fieldtarget=edition]}
3511 \map{step[fieldsource=paginas,fieldtarget=pages]}
3512 \map{step[fieldsource=numero,fieldtarget=number]}
3513 \map{step[fieldsource=tradutor,fieldtarget=translator]}
3514 \map{step[fieldsource=organizador,fieldtarget=editor]}
3515 \map{step[fieldsource=instituicao,fieldtarget=institution]}
3516 \map{step[fieldsource=tipo,fieldtarget=type]}
3517 \map{step[fieldsource=nota,fieldtarget=note]}
3518 \map{step[fieldsource=dataacesso,fieldtarget=urldate]}
3519 \map{step[fieldsource=curso,fieldtarget=course]}
3520 \map{step[fieldsource=grau,fieldtarget=coppedegree]}
3521 \map{step[fieldsource=artista,fieldtarget=artist]}
3522 \map{step[fieldsource=plataforma,fieldtarget=platform]}
3523 \map{step[fieldsource=diretor,fieldtarget=director]}
3524 \map{step[fieldsource=produtor,fieldtarget=producer]}
3525 \map{step[fieldsource=versao,fieldtarget=version]}
3526 \map{step[fieldsource=escala,fieldtarget=scale]}
3527 }%
3528 }
3529
3530 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
3531 \setlength{\bibhang}{0pt}
3532 \setlength{\bibitemsep}{\baselineskip}
3533
3534 </bbx>
3535 <*cbx>
3536 \ProvidesFile{coppe.cbx}[2026/05/28 v4.0 CoppeTeX ABNT author-date citations]
3537 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
3538 % author-year cites with normal-caps names (post-2020 rule) out of the box.
3539 \RequireCitationStyle{authoryear-comp}

```

```

3540
3541 % --- NBR 10520:2023 author separator -----
3542 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
3543 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
3544 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
3545 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
3546 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
3547 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
3548 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
3549 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
3550
3551 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
3552 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
3553 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
3554 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
3555 % so it is never added to the bibliography.
3556 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
3557 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
3558 % \citeapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
3559 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
3560 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
3561 \DeclareRobustCommand{\citetapud}[4][]{%
3562   #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
3563 \DeclareRobustCommand{\citeapud}[4][]{%
3564   \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
3565     \citeauthor{#4}\addcomma\space\citeyear{#4}%
3566     \ifblank{#1}{ }\addcomma\space\mkpageprefix[pagination]{#1}}}%
3567 \

```

```

3594 urlseen      = {{Access on}{Access on}},
3595 in            = {{In}{In}},
3596 editor        = {{editor}{ed\addot}},
3597 editors       = {{editors}{ed\addot}},
3598 depositor     = {{Depositor}{Depositor}},
3599 coppescale    = {{Scale}{Scale}},
3600 mscdiss       = {{M\addot Sc\addot Dissertation}{M\addot Sc\addot Diss\addot}},
3601 dscthesi      = {{D\addot Sc\addot Thesis}{D\addot Sc\addot Thesis}},
3602 phdthesis     = {{Ph\addot D\addot Thesis}{Ph\addot D\addot Thesis}},
3603 tcc           = {{Undergraduate Thesis}{Undergraduate Thesis}},
3604 }
3605 </lboxen>
3606 <*lboxes>
3607 \ProvidesFile{spanish-coppe.lbx}[2026/05/28 v4.0 CoppeTeX spanish strings]
3608 \InheritBibliographyExtras{spanish}
3609 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
3610 \DeclareBibliographyStrings{%
3611   inherit      = {spanish},
3612   availablefrom = {{Disponible en}{Disponible en}},
3613   urlseen      = {{Consultado el}{Consultado el}},
3614   in           = {{En}{En}},
3615   editor       = {{editor}{ed\addot}},
3616   editors      = {{editores}{ed\addot}},
3617   depositor    = {{Solicitante}{Solicitante}},
3618   coppescale   = {{Escala}{Escala}},
3619   mscdiss      = {{Tesis de Maestr'\i a}{Tesis de Maestr'\i a}},
3620   dscthesi     = {{Tesis Doctoral}{Tesis Doctoral}},
3621   phdthesis    = {{Tesis Doctoral}{Tesis Doctoral}},
3622   tcc          = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
3623 }
3624 </lboxes>
3625 <*lboxfr>
3626 \ProvidesFile{french-coppe.lbx}[2026/05/28 v4.0 CoppeTeX french strings]
3627 \InheritBibliographyExtras{french}
3628 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
3629 \DeclareBibliographyStrings{%
3630   inherit      = {french},
3631   availablefrom = {{Disponible \a}{Disponible \a}},
3632   urlseen      = {{Consult\ 'e le}{Consult\ 'e le}},
3633   in           = {{Dans}{Dans}},
3634   editor       = {{\ 'editeur}{\ 'ed\addot}},
3635   editors      = {{\ 'editeurs}{\ 'ed\addot}},
3636   depositor    = {{D\ 'eposant}{D\ 'eposant}},
3637   coppescale   = {{\ 'Echelle}{\ 'Echelle}},
3638   mscdiss      = {{M\ 'emoire de Master}{M\ 'emoire de Master}},
3639   dscthesi     = {{Th\ 'ese de Doctorat}{Th\ 'ese de Doctorat}},
3640   phdthesis    = {{Th\ 'ese de Doctorat}{Th\ 'ese de Doctorat}},
3641   tcc          = {{M\ 'emoire de fin d\ 'etudes}{M\ 'emoire de fin d\ 'etudes}},
3642 }
3643 </lboxfr>
3644 <*lboxit>
3645 \ProvidesFile{italian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX italian strings]
3646 \InheritBibliographyExtras{italian}
3647 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}

```

```

3648 \DeclareBibliographyStrings{%
3649   inherit      = {italian},
3650   availablefrom = {{Disponibile da}{Disponibile da}},
3651   urlseen      = {{Consultato il}{Consultato il}},
3652   in           = {{In}{In}},
3653   editor       = {{curatore}{cur\adddot}},
3654   editors      = {{curatori}{cur\adddot}},
3655   depositor    = {{Depositante}{Depositante}},
3656   coppescale   = {{Scala}{Scala}},
3657   msdiss       = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
3658   dscthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
3659   phdthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
3660   tcc          = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
3661 }
3662 \ltxit
3663 <*\langes>
3664 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.
3665 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
3666 %% spanish slot) is active. Populates the dispatcher table for the
3667 %% spanish language and registers the biblatex language mapping.
3668 %% Institutional names (universityname/cityname/statename/countryname)
3669 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
3670 %% rosto are a Brazilian institutional template that does not translate.
3671 \ProvidesFile{coppe-lang-spanish.def}[2026/05/28 v4.0 CoppeTeX spanish class strings]
3672 \copperdefstring{spanish}{appendix}      {Ap\`endice}
3673 \copperdefstring{spanish}{annex}         {Anexo}
3674 \copperdefstring{spanish}{frame}         {Cuadro}
3675 \copperdefstring{spanish}{listframe}     {Lista de Cuadros}
3676 \copperdefstring{spanish}{source}        {Fuente}
3677 \copperdefstring{spanish}{program}       {Programa}
3678 \copperdefstring{spanish}{listprogram}   {Lista de Programas}
3679 \copperdefstring{spanish}{references}     {Referencias}
3680 \copperdefstring{spanish}{in:}          {en}
3681 \copperdefstring{spanish}{art}           {Arte}
3682 \copperdefstring{spanish}{platform}      {Plataforma}
3683 \copperdefstring{spanish}{directedby}    {Direcci\`on}
3684 \copperdefstring{spanish}{producedby}    {Producci\`on}
3685 \copperdefstring{spanish}{listabbreviation}{Lista de Abreviaturas y Siglas}
3686 \copperdefstring{spanish}{listsymbol}    {Lista de S\`imbolos}
3687 \copperdefstring{spanish}{glossary}      {Glosario}
3688 \copperdefstring{spanish}{advisor}       {Director}
3689 \copperdefstring{spanish}{advisors}      {Directores}
3690 \copperdefstring{spanish}{dept}          {Programa}
3691 \copperdefstring{spanish}{approvedmale}  {Aprobado por}
3692 \copperdefstring{spanish}{approvedfemale}{Aprobada por}
3693 \copperdefstring{spanish}{approvedonmale}{Aprobado el}
3694 \copperdefstring{spanish}{approvedonfemale}{Aprobada el}
3695 \copperdefstring{spanish}{keywordsname}  {Palabras clave}
3696 \copperdefstring{spanish}{volumename}    {Volumen}
3697 \copperdefstring{spanish}{coadvisor}     {Codirector}
3698 \copperdefstring{spanish}{coadvisors}    {Codirectores}
3699 \copperdefstring{spanish}{ofname}        {de}
3700 \copperdefstring{spanish}{universityname}{Universidade Federal do Rio de Janeiro}
3701 \copperdefstring{spanish}{cityname}     {Rio de Janeiro}

```

```

3702 \copperdefstring{spanish}{statename} {RJ}
3703 \copperdefstring{spanish}{countryname} {Brasil}
3704 \copperdefstring{spanish}{monthname1} {enero}
3705 \copperdefstring{spanish}{monthname2} {febrero}
3706 \copperdefstring{spanish}{monthname3} {marzo}
3707 \copperdefstring{spanish}{monthname4} {abril}
3708 \copperdefstring{spanish}{monthname5} {mayo}
3709 \copperdefstring{spanish}{monthname6} {junio}
3710 \copperdefstring{spanish}{monthname7} {julio}
3711 \copperdefstring{spanish}{monthname8} {agosto}
3712 \copperdefstring{spanish}{monthname9} {septiembre}
3713 \copperdefstring{spanish}{monthname10} {octubre}
3714 \copperdefstring{spanish}{monthname11} {noviembre}
3715 \copperdefstring{spanish}{monthname12} {diciembre}
3716 \copperdefstring{spanish}{algorithm2eopt}{spanish}
3717 \copperdefstring{spanish}{babelname} {spanish}
3718 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
3719 %% language-mapping registration until \begin{document} so biblatex's
3720 %% \DeclareLanguageMapping is available.
3721 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
3722 \end{langes}
3723 \end{*langfr}
3724 %% coppe-lang-french.def -- CoppeTeX 4.x French class-string pack.
3725 %% See coppe-lang-spanish.def for the structure. Institutional names
3726 %% remain Portuguese; only document-content strings translate.
3727 \ProvidesFile{coppe-lang-french.def}[2026/05/28 v4.0 CoppeTeX french class strings]
3728 \copperdefstring{french}{appendix} {Annexe}
3729 \copperdefstring{french}{annex} {Annexe}
3730 \copperdefstring{french}{frame} {Encadr\'e}
3731 \copperdefstring{french}{listframe} {Liste des encadr\'es}
3732 \copperdefstring{french}{source} {Source}
3733 \copperdefstring{french}{program} {Programme}
3734 \copperdefstring{french}{listprogram} {Liste des programmes}
3735 \copperdefstring{french}{references} {R\'ef\'erences}
3736 \copperdefstring{french}{in:} {dans}
3737 \copperdefstring{french}{art} {Illustration}
3738 \copperdefstring{french}{platform} {Plateforme}
3739 \copperdefstring{french}{directedby} {R\'ealis\'e par}
3740 \copperdefstring{french}{producedby} {Produit par}
3741 \copperdefstring{french}{listabbreviation} {Liste des abr\'eviations et acronymes}
3742 \copperdefstring{french}{listsymbol} {Liste des symboles}
3743 \copperdefstring{french}{glossary} {Glossaire}
3744 \copperdefstring{french}{advisor} {Directeur}
3745 \copperdefstring{french}{advisors} {Directeurs}
3746 \copperdefstring{french}{dept} {Programme}
3747 \copperdefstring{french}{approvedmale} {Approuv\'e par}
3748 \copperdefstring{french}{approvedfemale} {Approuv\'ee par}
3749 \copperdefstring{french}{approvedonmale} {Approuv\'e le}
3750 \copperdefstring{french}{approvedonfemale} {Approuv\'ee le}
3751 \copperdefstring{french}{keywordsname} {Mots-cl\'es}
3752 \copperdefstring{french}{volumename} {Volume}
3753 \copperdefstring{french}{coadvisor} {Co-directeur}
3754 \copperdefstring{french}{coadvisors} {Co-directeurs}
3755 \copperdefstring{french}{ofname} {de}

```

```

3756 \copperdefstring{french}{universityname}{Universidade Federal do Rio de Janeiro}
3757 \copperdefstring{french}{cityname} {Rio de Janeiro}
3758 \copperdefstring{french}{statename} {RJ}
3759 \copperdefstring{french}{countryname} {Brasil}
3760 \copperdefstring{french}{monthname1} {janvier}
3761 \copperdefstring{french}{monthname2} {f\'evrier}
3762 \copperdefstring{french}{monthname3} {mars}
3763 \copperdefstring{french}{monthname4} {avril}
3764 \copperdefstring{french}{monthname5} {mai}
3765 \copperdefstring{french}{monthname6} {juin}
3766 \copperdefstring{french}{monthname7} {juillet}
3767 \copperdefstring{french}{monthname8} {ao\^ut}
3768 \copperdefstring{french}{monthname9} {septembre}
3769 \copperdefstring{french}{monthname10}{octobre}
3770 \copperdefstring{french}{monthname11}{novembre}
3771 \copperdefstring{french}{monthname12}{d\'ecembre}
3772 \copperdefstring{french}{algorithm2eopt}{french}
3773 \copperdefstring{french}{babelname} {french}
3774 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
3775 \end{langfr}
3776 \end{langit}
3777 %% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
3778 %% See coppe-lang-spanish.def for the structure. The algorithm2e
3779 %% language option for Italian is spelled "italiano" -- that's the
3780 %% spelling algorithm2e ships, even though babel calls the language
3781 %% "italian".
3782 \ProvidesFile{coppe-lang-italian.def}[2026/05/28 v4.0 CoppeTeX italian class strings]
3783 \copperdefstring{italian}{appendix} {Appendice}
3784 \copperdefstring{italian}{annex} {Allegato}
3785 \copperdefstring{italian}{frame} {Riquadro}
3786 \copperdefstring{italian}{listframe} {Elenco dei riquadri}
3787 \copperdefstring{italian}{source} {Fonte}
3788 \copperdefstring{italian}{program} {Programma}
3789 \copperdefstring{italian}{listprogram} {Elenco dei programmi}
3790 \copperdefstring{italian}{references} {Riferimenti}
3791 \copperdefstring{italian}{in:} {in}
3792 \copperdefstring{italian}{art} {Illustrazione}
3793 \copperdefstring{italian}{platform} {Piattaforma}
3794 \copperdefstring{italian}{directedby} {Regia di}
3795 \copperdefstring{italian}{producedby} {Prodotto da}
3796 \copperdefstring{italian}{listabbreviation}{Elenco di abbreviazioni e sigle}
3797 \copperdefstring{italian}{listsymbol} {Elenco dei simboli}
3798 \copperdefstring{italian}{glossary} {Glossario}
3799 \copperdefstring{italian}{advisor} {Relatore}
3800 \copperdefstring{italian}{advisors} {Relatori}
3801 \copperdefstring{italian}{dept} {Programma}
3802 \copperdefstring{italian}{approvedmale} {Approvato da}
3803 \copperdefstring{italian}{approvedfemale} {Approvata da}
3804 \copperdefstring{italian}{approvedonmale} {Approvato il}
3805 \copperdefstring{italian}{approvedonfemale}{Approvata il}
3806 \copperdefstring{italian}{keywordsname} {Parole chiave}
3807 \copperdefstring{italian}{volumename} {Volume}
3808 \copperdefstring{italian}{coadvisor} {Correlatore}
3809 \copperdefstring{italian}{coadvisors} {Correlatori}

```



```

3810 \copperdefstring{italian}{ofname}      {di}
3811 \copperdefstring{italian}{universityname}{Universidade Federal do Rio de Janeiro}
3812 \copperdefstring{italian}{cityname}     {Rio de Janeiro}
3813 \copperdefstring{italian}{statename}    {RJ}
3814 \copperdefstring{italian}{countryname}  {Brasil}
3815 \copperdefstring{italian}{monthname1}   {gennaio}
3816 \copperdefstring{italian}{monthname2}   {febbraio}
3817 \copperdefstring{italian}{monthname3}   {marzo}
3818 \copperdefstring{italian}{monthname4}   {aprile}
3819 \copperdefstring{italian}{monthname5}   {maggio}
3820 \copperdefstring{italian}{monthname6}   {giugno}
3821 \copperdefstring{italian}{monthname7}   {luglio}
3822 \copperdefstring{italian}{monthname8}   {agosto}
3823 \copperdefstring{italian}{monthname9}   {settembre}
3824 \copperdefstring{italian}{monthname10}  {ottobre}
3825 \copperdefstring{italian}{monthname11} {novembre}
3826 \copperdefstring{italian}{monthname12} {dicembre}
3827 \copperdefstring{italian}{algorithm2eopt}{italiano}
3828 \copperdefstring{italian}{babelname}    {italian}
3829 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
3830 </langit>
3831 <*numbbx>
3832 \ProvidesFile{coppe-numeric.bbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric bibliography]
3833 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
3834 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
3835 % the PT->EN source map and the language mappings) and only adds the [n] list
3836 % label and the numeric sorting on top.
3837 \RequireBibliographyStyle{coppe}
3838
3839 % ABNT numeric system lists references in the order they are first cited
3840 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
3841 % numbered list. 'labelnumber' generates the [n] field.
3842 \ExecuteBibliographyOptions{labelnumber,sorting=none}
3843
3844 % Bracketed labels, both in the list and for any shorthand entries.
3845 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
3846 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
3847
3848 % Reference list driven by the [n] label (copied from biblalex's numeric.bbx),
3849 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
3850 \defbibenvironment{bibliography}
3851 {
3852   \list
3853     {\printtext[labelnumberwidth]{%
3854       \printfield{labelprefix}%
3855       \printfield{labelnumber}}}
3856     {\setlength{\labelwidth}{\labelnumberwidth}%
3857      \setlength{\leftmargin}{\labelwidth}%
3858      \setlength{\labelsep}{\biblabelsep}%
3859      \addtolength{\leftmargin}{\labelsep}%
3860      \setlength{\itemsep}{\bibitemsep}%
3861      \setlength{\parsep}{\bibparsep}}%
3862     {\renewcommand*{\makelabel}[1]{\hss#1}}
3863 }

```

```

3864
3865 </numbbx>
3866 <*numcbx>
3867 \ProvidesFile{coppe-numeric.cbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric citations]
3868 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
3869 % numbers [1], compressed and sorted within a single cite group. This is the
3870 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
3871 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
3872 % \citet -> Author [1].
3873 \RequireCitationStyle{numeric-comp}
3874 </numcbx>

```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

10.3 The generated companion files

Everything the distribution ships is extracted from this file. The class, the BibLaTeX styles, the language packs, the databases and the `.ist` were already here; the demonstration documents, the `latexmkrc` and the regression tests joined them, so that a single `pdflatex coppe.ins` rebuilds the whole tree and there is no second place to edit. Each one sits in an `\iffalse` block: the documentation driver never typesets them, and only `coppe.ins` extracts them.

The tests live in `src/tests/` rather than a sibling `tests/`. That is not taste: $\TeX$ refuses `\openout` on a path containing `..`, so a `.ins` run in `src/` can write into a subdirectory and never into a parent. Putting the suite under `src/` is what lets it be generated at all.

What is NOT generated from here, and why: `tools/*.ps1` and `src/tests/run-tests.ps1` are the build harness, not artefacts of the class; `NORMA_COPPE_2026.tex` and `futuremanual2026.tex` are documents about the norm that happen to live in this repository, with their own life cycle.

10.3.1 `example_pt.tex` -- pt-main demo

`example_pt.tex` – CoppeTeX 4.0 multilingual demo, main language: brazilian. Short demonstration document (cover + folha + abstracts + 1 chapter) used by the CPGP submission package. The full pt-main example with bibliography and lists is the historical `example.tex`.

10.3.2 `example_en.tex` -- en-main demo

`example_en.tex` – CoppeTeX 4.0 multilingual demo, main language: english.

10.3.3 `example_es.tex` -- es-main demo

`example_es.tex` – CoppeTeX 4.0 multilingual demo, main language: spanish. Requires the language pack files `coppe-lang-spanish.def` and `spanish-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.4 `example_fr.tex` -- fr-main demo

`example_fr.tex` – CoppeTeX 4.0 multilingual demo, main language: french. Requires the language pack files `coppe-lang-french.def` and `french-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.5 `example_it.tex` -- it-main demo

`example_it.tex` – CoppeTeX 4.0 multilingual demo, main language: italian. Requires the language pack files `coppe-lang-italian.def` and `italian-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.6 `example_pdfa.tex` -- the full example as PDF/A

`example_pdfa.tex` – o `example.tex` completo, compilado com a opção ‘pdfa’.

Existe para dar ao veraPDF o documento mais exigente que a classe produz: o exemplo inteiro, com bibliografia, listas, figuras, tabelas, algoritmos e as caixas do `tcolorbox` – ou seja, transparência, que é justamente o que o a-1b proibiria e o a-2b admite. Validar só um documento curto não diria nada sobre o que vai de fato para o depósito.

`\PassOptionsToClass` tem de vir ANTES do `\documentclass` que o `example.tex` traz dentro de si; é o único jeito de acrescentar uma opção de classe a um arquivo gerado pelo `docstrip` sem duplicá-lo aqui e deixar as duas cópias divergirem com o tempo.

10.3.7 `covers_5languages.tex` -- the five covers side by side

`covers_5languages.tex` – One-page visual demonstration of the CoppeTeX 4.0 multilingual model: the five built-in main languages side by side, all sharing the same Portuguese institutional template, differing only in the title (and the body language, not visible on the cover).

This file uses the article class deliberately – it is a meta-document about the coppe class, not a thesis. It expects the five cover PNGs `cover_pt.png`, `cover_en.png`, `cover_es.png`, `cover_fr.png`, `cover_it.png` to live in the same directory (generated by `pdftoppm` from the `example_<lang>.pdf` demos).

10.3.8 `latexmkrc` -- latexmk/Overleaf build configuration

`latexmkrc` – CoppeTeX build configuration (works locally and on Overleaf).

Run with: `latexmk example` (or `latexmk futuremanual2026`, your thesis...) `latexmk` reads this file automatically when it is in the build directory. On Overleaf, place this file at the project root; Overleaf runs `latexmk`.

Engine: pdfLaTeX by default (fast). For LuaLaTeX (recommended, full Unicode) set `$pdf_mode = 4` below, or use the `"%!TeX program = lualatex"` magic comment / the Overleaf "Menu > Compiler" setting.

10.3.9 tests/test_brazilian_one_advisor.tex

% % test_brazilian_one_advisor.tex – regression guard. % % Historical bug: when only ONE \advisor was given, the cover/folha-de-rosto % rendering used to fall over (the original \maketitle loop assumed at % least two advisors). This test pins the single-advisor path: it must % compile cleanly AND the cover label must read "Orientador" (singular), % not "Orientadores". % % Scope: just enough thesis to render \maketitle and the front pages. % No body chapters – they are covered by test_brazilian_two_advisors.tex. % % Run via ../tests/run-tests.ps1. %

10.3.10 tests/test_brazilian_two_advisors.tex

% % test_brazilian_two_advisors.tex – regression test for the coppe class. % Brazilian main language, two advisors (the common case; the cover label % should print "Orientadores"). % % Run via ../tests/run-tests.ps1, which sets TEXINPUTS so this file finds % coppe.cls and example.bib in ../src/. EXTEND THIS FILE whenever a new % bug is found in the two-advisor / common path: keep every existing case % working and add a minimal repro for the new one. % % Covered features (as of 2026/05/29, CoppeTeX v4.0, nlinguas branch): % * metadata: \title, \foreigntitle, \author, \advisor (x2), \examiner (x3), % \department, \date, \keyword (x3) % * \maketitle (cover + folha de rosto + ficha catalografica) % * \frontmatter: \dedication, agradecimentos chapter*, abstract, foreignabstract % * lists: \listoffigures, \listoftables, \listofquadros, \listofprogramas, % \printloabbreviations, \printlosymbols, \tableofcontents % * \mainmatter chapters covering: figures with \source/\fonte, booktabs % table, quadro float, subcaption subfigures, longtable, amsmath + % amssymb math, algorithm2e algorithm, listings Python block, longquote, % footnotes (singlespacing+filete via setspace), \sigla / \sigla*, % \syml, \abbrev, \citet, \citep, \autoref cross-references % * back matter: \printbibliography, \appendix (x2), \annex % % Compile under pdfLaTeX. Under LuaLaTeX/XeLaTeX, the class skips amssymb % and the end-of-document ClassWarning will fire (test it by adding % \usepackage{unicode-math} + \setmathfont{Latin Modern Math} below). %

10.3.11 tests/test_brazilian_three_advisors.tex

% % test_brazilian_three_advisors.tex – upper-bound regression test. % % Three \advisor calls is the most the COPPE/UFRJ cover is expected to % typeset reasonably; the class must still render the cover and folha de % rosto cleanly, with the plural label "Orientadores". % % Scope: just enough thesis to render \maketitle and the front pages. % No body chapters – they are covered by test_brazilian_two_advisors.tex. % % Run via ../tests/run-tests.ps1. %

10.3.12 tests/test_assinaturas.tex

% % test_assinaturas.tex – regression test for the coppe class. % Exercises the pre-textual pages introduced/reshaped for the SiBI manual, % 9th revised edition (2026): % % * class option 'assinaturas' – folha de aprovacao with one signature % rule per board member instead of the compact list (Annex D) % * \examiner and \advisor with the OPTIONAL institution argument, and the % titulacao argument that pre-4.1 discarded silently % * folha adicional with the Coleta CAPES fields

(3.1.2.1.2, Annex H): % \tipoproducao, \projetovinculado, \nomeprojeto, \areaconcentracao, % \agenciafomento % * \linhapesquisa and \areaconcentracao on the folha de rosto % * \dataaprovacao on the folha de aprovacao % % Deliberately mixes calls WITH and WITHOUT the optional institution, so a % regression in the optional-argument handling shows up here. % % Run via ../tests/run-tests.ps1. Compile under pdfLaTeX. %

10.3.13 tests/test_assinaturas_7.tex

% % test_assinaturas_7.tex – regression test for the coppe class. % % The folha de aprovacao under the ‘assinaturas’ option gives every board % member a signature rule of its own (Annex D). The class had only ever been % exercised up to five members, and a doctoral board at the COPPE routinely % reaches seven, so this test pins the tall end of the range: two advisors % plus five examiners, one of them with a long name and a long institution, % which is the combination most likely to wrap a line and push the sheet over. % % What "passing" means here is narrower than "pdflatex exited 0": the folha % de aprovacao must still fit on ONE page. tools/check-aprovacao.py measures % that from the rendered PDF; run-tests.ps1 only sees the exit code, so a % silent overflow would show up as an Overfull \vbox in the log rather than % as a failure. Both are checked. % % Run via ../tests/run-tests.ps1. Compile under pdfLaTeX. %

10.3.14 tests/test_pdfa.tex

% % test_pdfa.tex – regression test for the coppe class. % Exercises the pre-textual pages introduced/reshaped for the SiBI manual, % 9th revised edition (2026): % % * class option ‘assinaturas’ – folha de aprovacao with one signature % rule per board member instead of the compact list (Annex D) % * \examiner and \advisor with the OPTIONAL institution argument, and the % titulacao argument that pre-4.1 discarded silently % * folha adicional with the Coleta CAPES fields (3.1.2.1.2, Annex H): % \tipoproducao, \projetovinculado, \nomeprojeto, \areaconcentracao, % \agenciafomento % * \linhapesquisa and \areaconcentracao on the folha de rosto % * \dataaprovacao on the folha de aprovacao % % Deliberately mixes calls WITH and WITHOUT the optional institution, so a % regression in the optional-argument handling shows up here. % % Run via ../tests/run-tests.ps1. Compile under pdfLaTeX. %

Acknowledgments

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General: Unification of the code for the list of symbols and abbreviations.	1	General: Some examples for figures, tables, longtables, etc. .	1
v0.3		v3.4.1	
General: Added 'draft' option. . . .	1	General: Fix english option to consider brazilian a second language, describe it in document	47
<code>\maketitle</code> : Added number of examiners test.	57	v3.5	
Generalization.	57	General: New command <code>annex</code> . .	93
v0.4		v3.5.1	
General: Beta documentation. . . .	1	General: New command <code>annex</code> now supports brazilian or english .	93
v0.5		v4.0	
<code>abstract</code> : Changed from macro to environment.	64	General: Catalog-label writers (<code>coppe@mainBegin</code> , <code>coppe@bibBegin</code> , <code>coppe@bibEnd</code> , <code>coppe@hasLof</code>) made language-agnostic: <code>detokenize</code> wrap on label names protects ':' and '.' from babel shorthand activation, and the raw <code>etex</code> primitives <code>romannumeral/number</code> replace LaTeX's <code>roman/arabic</code> to bypass <code>spanish-babel</code> 's <code>small-caps @roman</code> redefinition that would otherwise crash <code>pdftex</code> with "Unbalanced write command" on every <code>es-main</code> document.	1
<code>\backmatter</code> : Added <code>mainmatter</code> pages counter.	53	Class-wide string dispatcher (<code>copperdefstring</code> , <code>coppestring</code> , <code>coppemainstring</code> , <code>coppforeignstring</code>) replaces scattered <code>iflanguage</code> and <code>if</code> english branches; <code>pt/en</code> outputs remain byte-identical. .	1
<code>foreignabstract</code> : Changed from macro to environment.	65	Multilingual model: three fixed slots (<code>main</code> , <code>foreign</code> , <code>optional</code> <code>third</code>); five built-in language options (<code>brazilian</code> , <code>english</code> ,	
v1.0			
General: First COPPETeX release. .	1		
v2.0			
General: COPPETeX release 2.0. . .	1		
v2.1			
General: COPPETeX release 2.1:			
Matching the new rules.	1		
<code>\advisor</code> : Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors. .	50		
v2.2			
General: Matching new guidelines, including new logo.	1		
v2.2.2			
General: Fixed some text constants in <code>.bib</code> and documented it here. .	1		
v3.0			
General: Added support for monographs in English.	47		
New approval page layout. . . .	59		
<code>\department</code> : Added new course on Nanotechnology.	48		
<code>\examiner</code> : Examiners expansion without degree.	50		

spanish, french, italian);	usecoppelanguage,
plug-in mechanism for any	brazilianabstract environment. . . 1
other Babel language via	
coppe-lang-lang.def plus	\coppetexfinalpage: Initial
lang-coppe.lbx. 1	coppetexfinalpage colophon
New user API: titlein,	page. 54